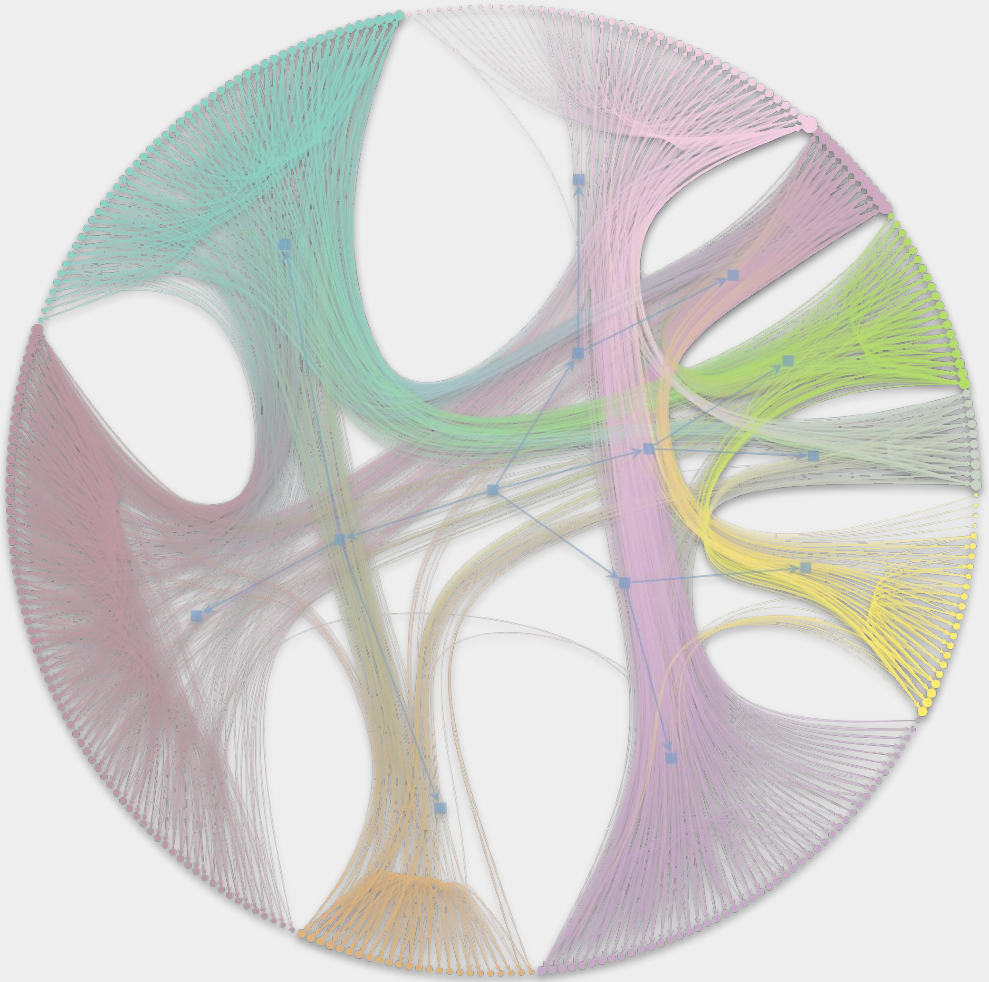
Network Elements

Part 01

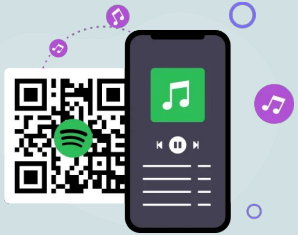
ivanovitch.silva@ufrn.br
@ivanovitchm

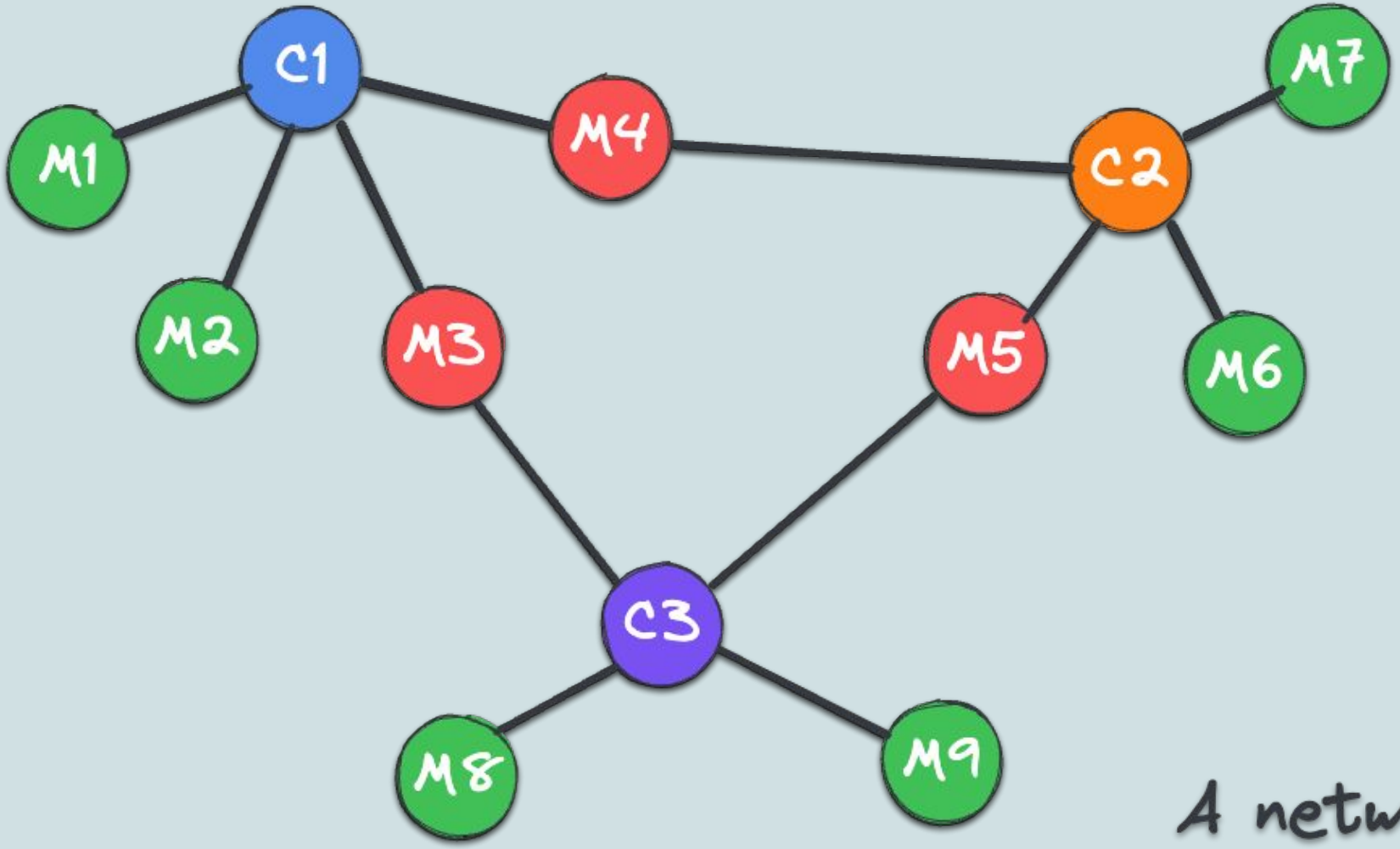
UFRN UNIVERSIDADE FEDERAL DO RIO GRANDE DO NORTE

DCA

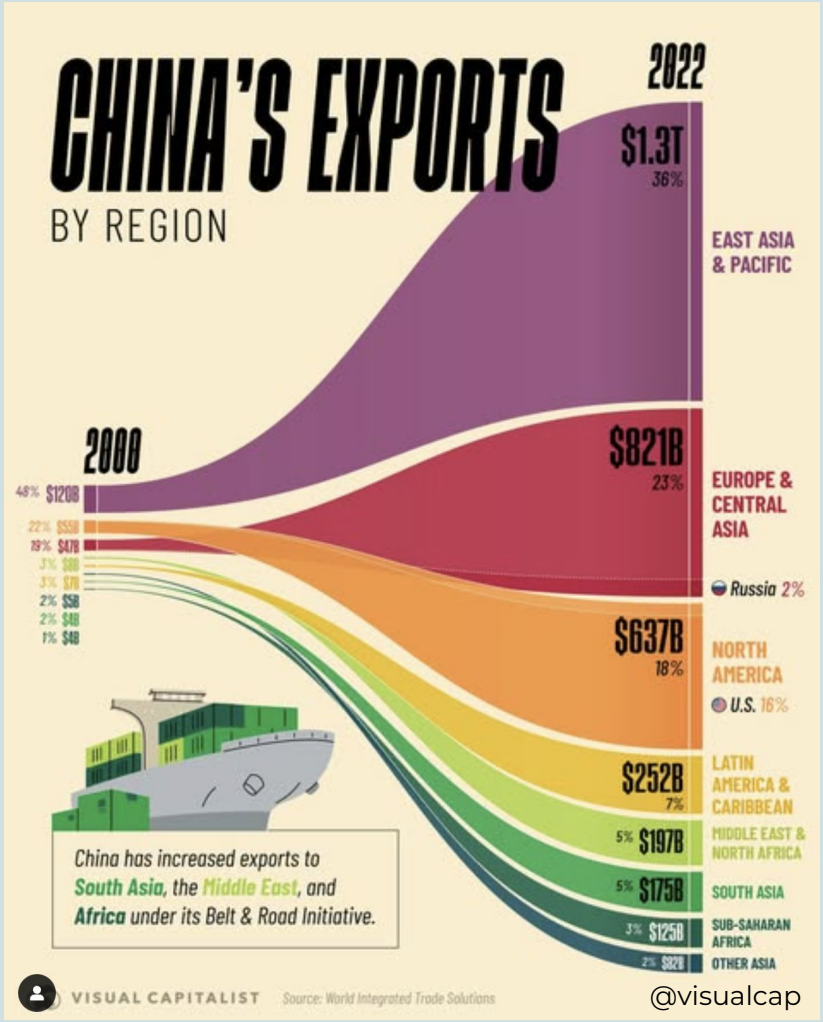




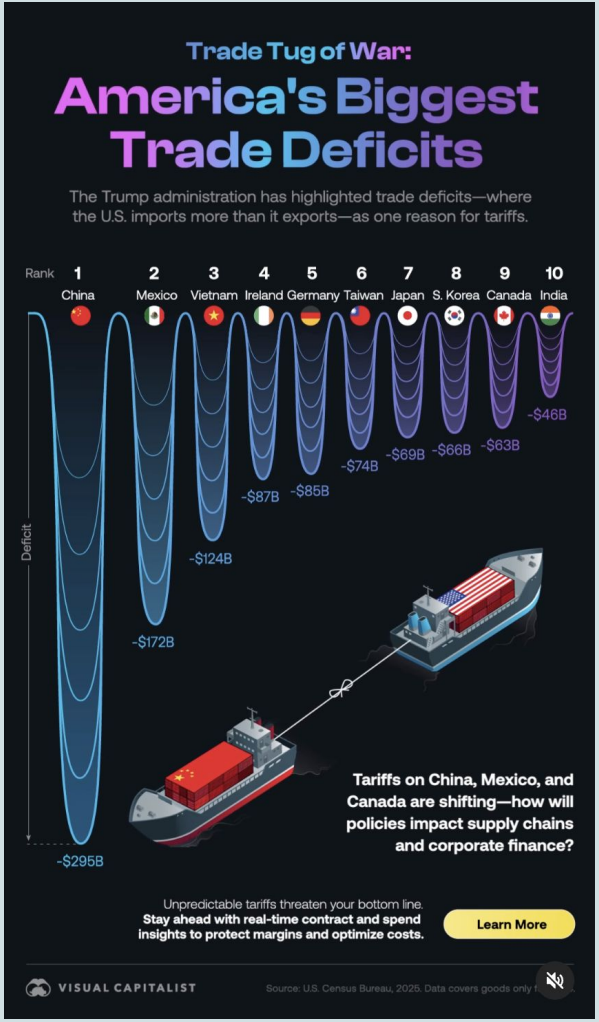




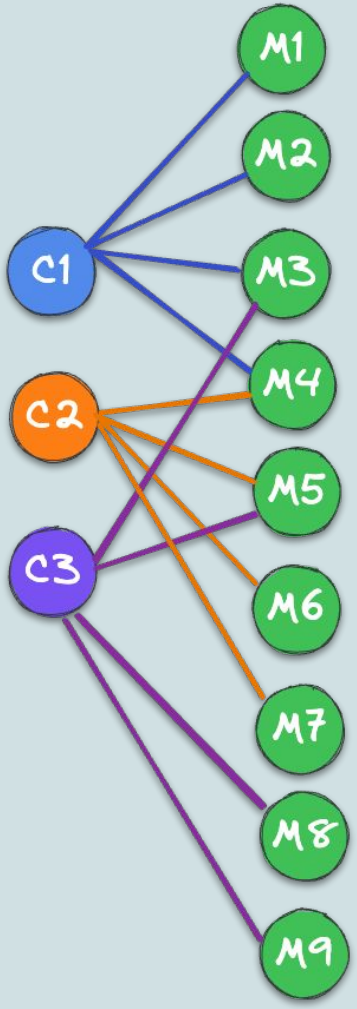
A network



@visualcap

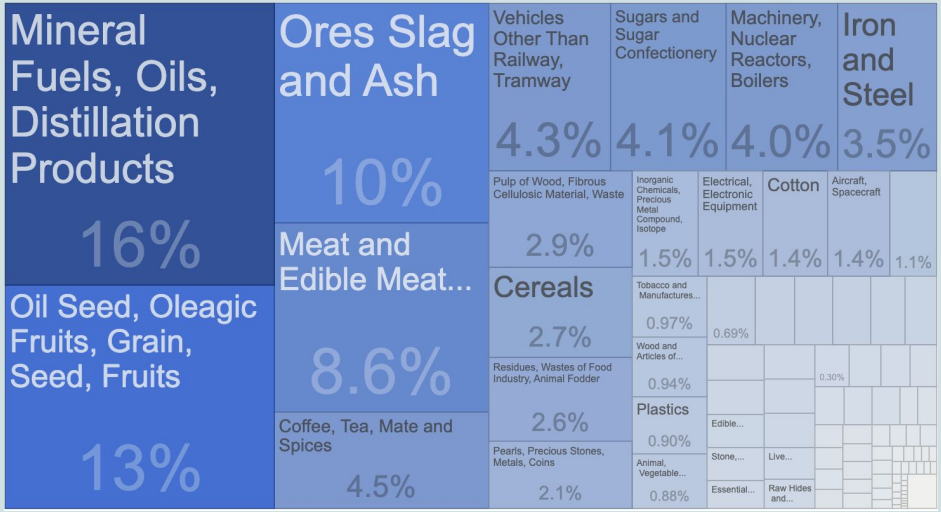


Network?

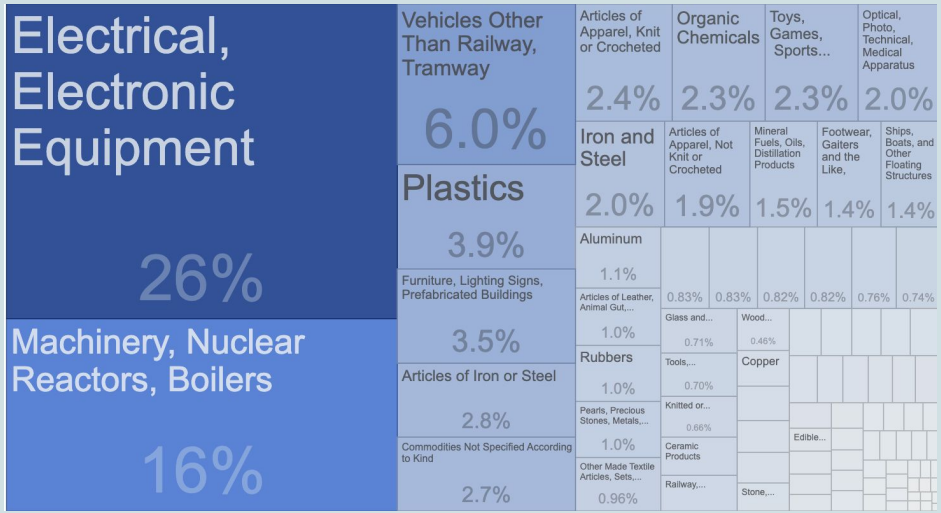


bipartite network



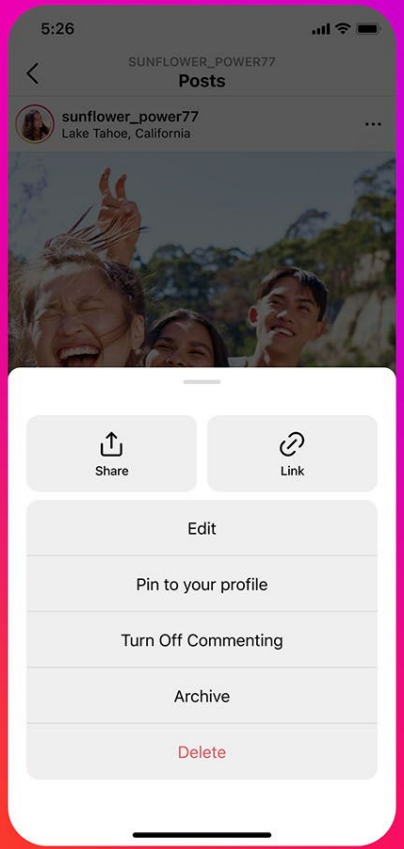


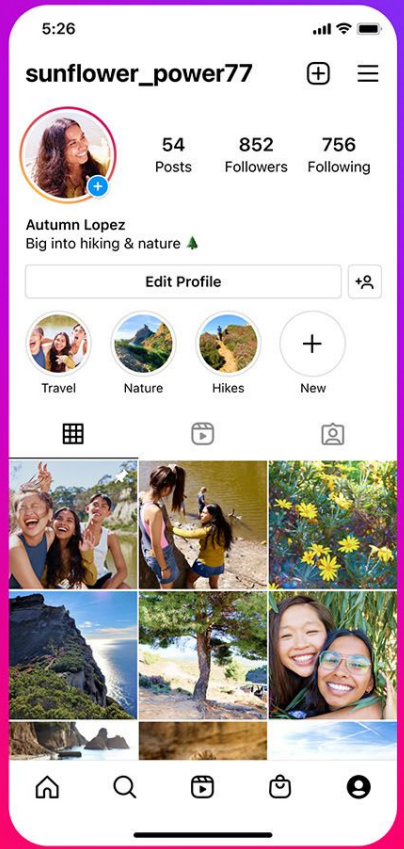


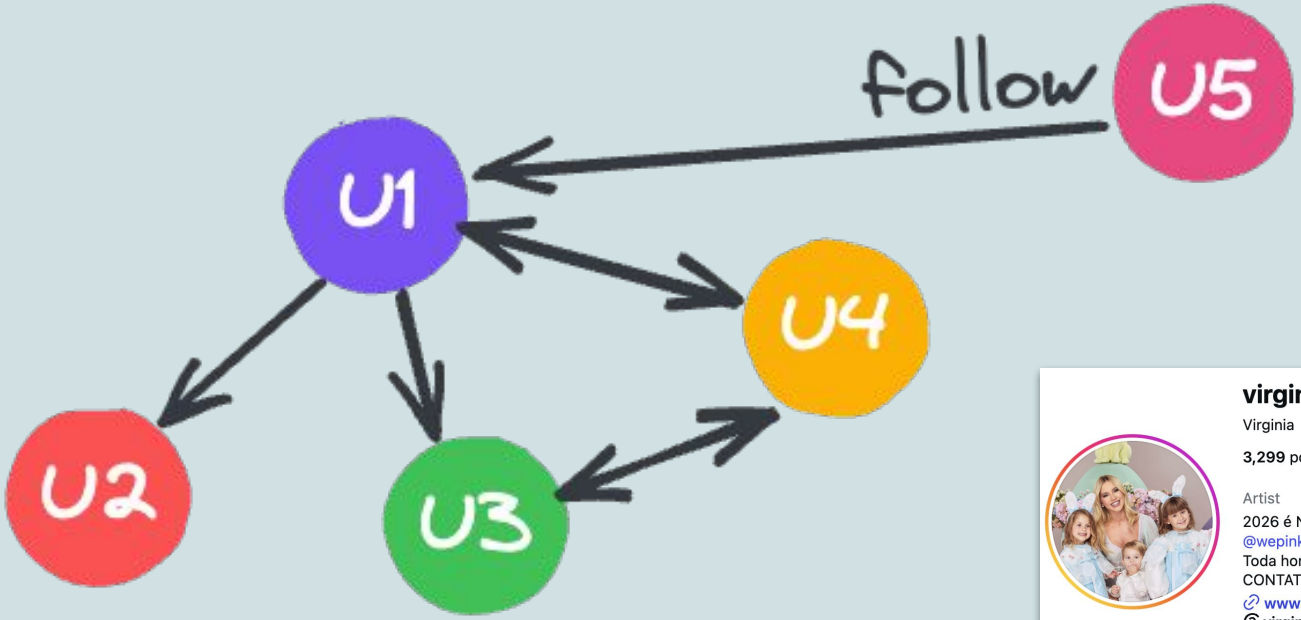


https://tradingeconomics.com/china/exports-by-category


5:26
SUNFLOWER_POWER77
Posts
sunflower_power77
Lake Tahoe, California
Now you can pin posts to your profile
Liked by liam_beanz99 and 12 others
sunflower_power77 Lake trip with the best ❤️
#laketahoe2022
View all 5 comments
Add a comment...
1 day ago


5:26
SUNFLOWER_POWER77
Posts
sunflower_power77
Lake Tahoe, California
Share
Link
Edit
Pin to your profile
Turn Off Commenting
Archive
Delete


5:26
sunflower_power77
54
Posts
852
Followers
756
Following
Autumn Lopez
Big into hiking & nature
Edit Profile
Travel
Nature
Hikes
New

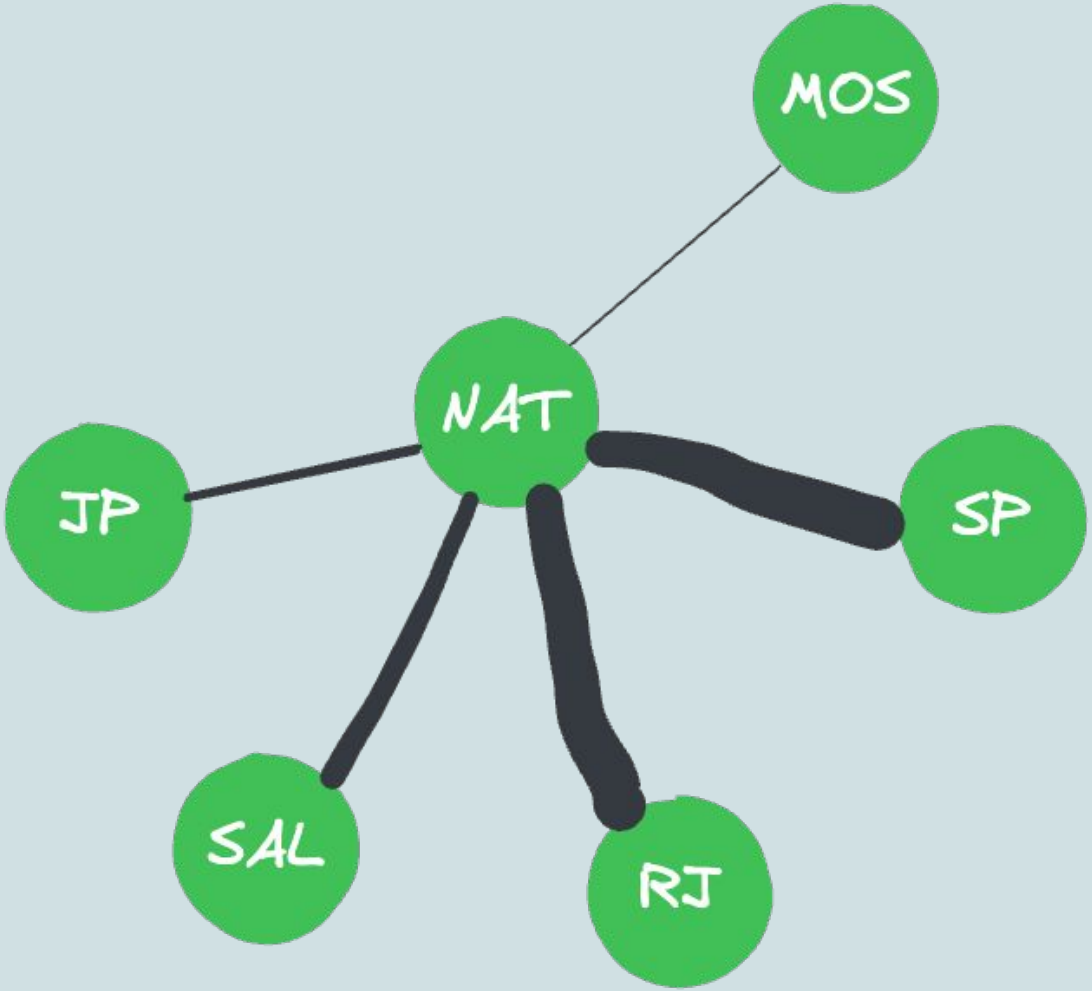

follow
U5
U1
U4
U2
U3

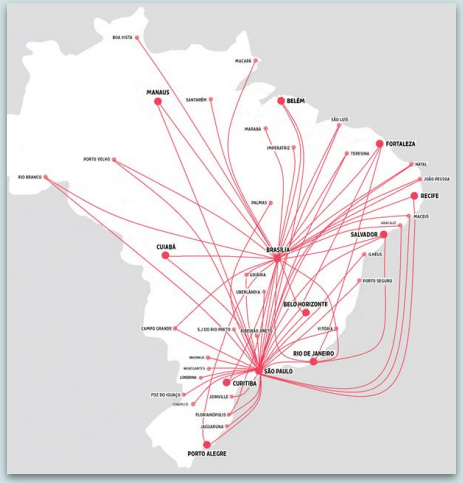

virginia
Virginia
3,299 posts 56.4M followers 735 following
Artist
2026 é NOSSO 💖💜💙
@wepink.br @wpink.br @casavfeventos @wegym_br
Toda honra e glória a Deus 🙌🏻
CONTATO:... more
www.wepink.com.br/?utm_source=virginia&utm_medium=instavi&ut...
virginia


leomessi
Leo Messi
1,535 posts 516M followers 369 following
Athlete
Bienvenidos a la cuenta oficial de Instagram de Leo Messi / Welcome to the official Leo Messi Instagram account
instagram.com/uecornella and 2 more




Most tracked flights LIVE
Airport disruptions LIVE
Bookmarks
flightradar24
LIVE AIR TRAFFIC
Find flights, airports and more
LOG IN
VIEW Map
Greenland
Iceland
Finland
Sweden
Norway
Russia
Ukraine
Mongolia
North Atlantic Ocean
North Pacific Ocean
Libya
Mali
Chad
Sudan
Iraq
Iran
Venezuela
Brazil
Bolivia
DRC
Namibia
Madagascar
South Africa
Papua New Guinea
South Pacific Ocean
South Atlantic Ocean
Indian Ocean
Southern Ocean
Settings
Weather
Filters
Widgets
Playback
Google
Keyboard shortcuts
Map Data ©2026
500 km
Terms


MOS
NAT
JP
SP
SAL
RJ


MANAUS
BELÉM
FORTALEZA
RECIFE
SALVADOR
CUIABÁ
BRASÍLIA
BELO HORIZONTE
RIO DE JANEIRO
SÃO PAULO
CURITIBA
PORTO ALEGRE

dataset-flights-brazil

Public

Watch 1 Fork 1 Starred 10

main 1 Branch 0 Tags Go to file t Add file Code

alvarofpp docs: Adds CDLA-Sharing-1.0 license aa8d01b · 6 months ago 39 Commits

.githooks	build: Adds Docker and Makefile	2 years ago
.github/workflows	ci: Adds integration workflow	2 years ago
data	chore: Updates data files	2 years ago
.gitignore	build: Adds Docker and Makefile	2 years ago
Dockerfile	build: Adds Docker and Makefile	2 years ago
LICENSE	docs: Adds LICENSE	2 years ago
Makefile	refact: Makefile	2 years ago
README.md	fix: Write the same file twice	2 years ago
docker-compose.yml	build: Adds Docker and Makefile	2 years ago
extract.py	style: Applies linters	2 years ago
requirements.txt	build: Adds Docker and Makefile	2 years ago
transform_to_airports_csv.py	style: Applies linters	2 years ago
transform_to_anac_csv.py	style: Applies linters	2 years ago
transform_to_graphml.py	style: Applies linters	2 years ago

About

Flights in Brazil registered by ANAC (Agência Nacional de Aviação Civil - National Civil Aviation Agency).

csv brazil python3 dataset flights air-traffic graphml air-traffic-data

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View license
Activity
10 stars
1 watching
1 fork
Report repository

Contributors 1

alvarofpp Álvaro Paiva

Languages

Python 86.3% Makefile 7.4% Dockerfile 3.7% Shell 2.6%

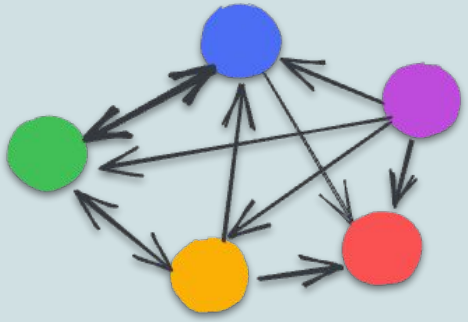




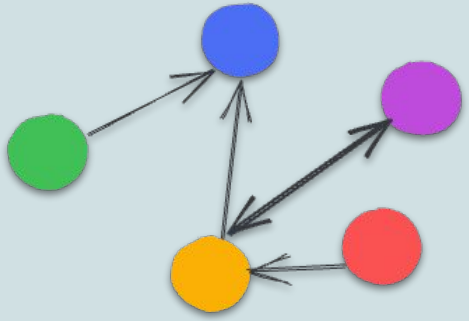


https://github.com/kurasaiteja/Whatsapp-Analysis

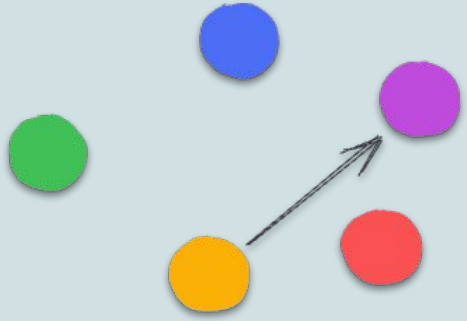
Monday



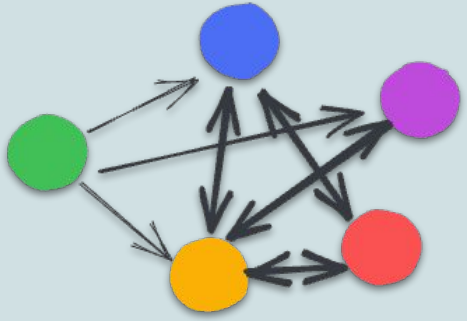
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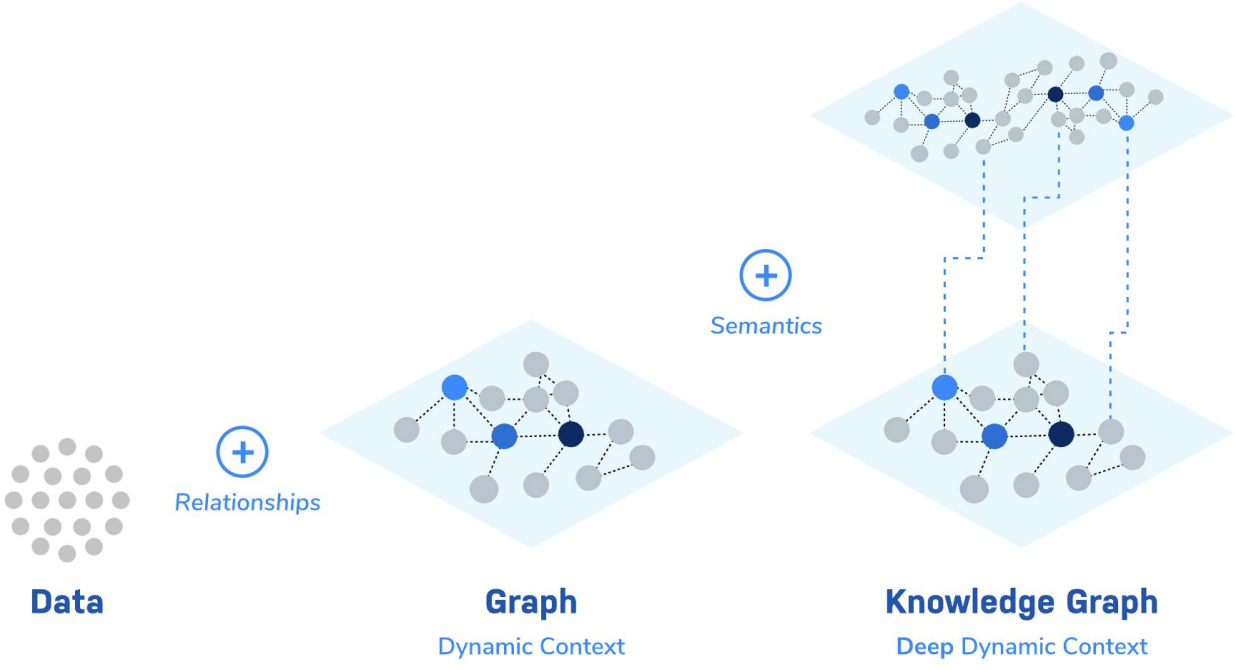
Wednesday



Thursday



Temporal network



1. Data

Bridge together diverse and disparate data silos regardless of data type, such as structured, unstructured, and semi-structured.

2. Graph

Map data and draw connections among them for the first layer of dynamic context, which provides immediate understanding.

3. Semantics

Apply semantics to provide deeper context to connected data. The deeper the context, the more powerful the insights.

MEM GRAPH

neo4j

Retrieval Augmented Generation (RAG)

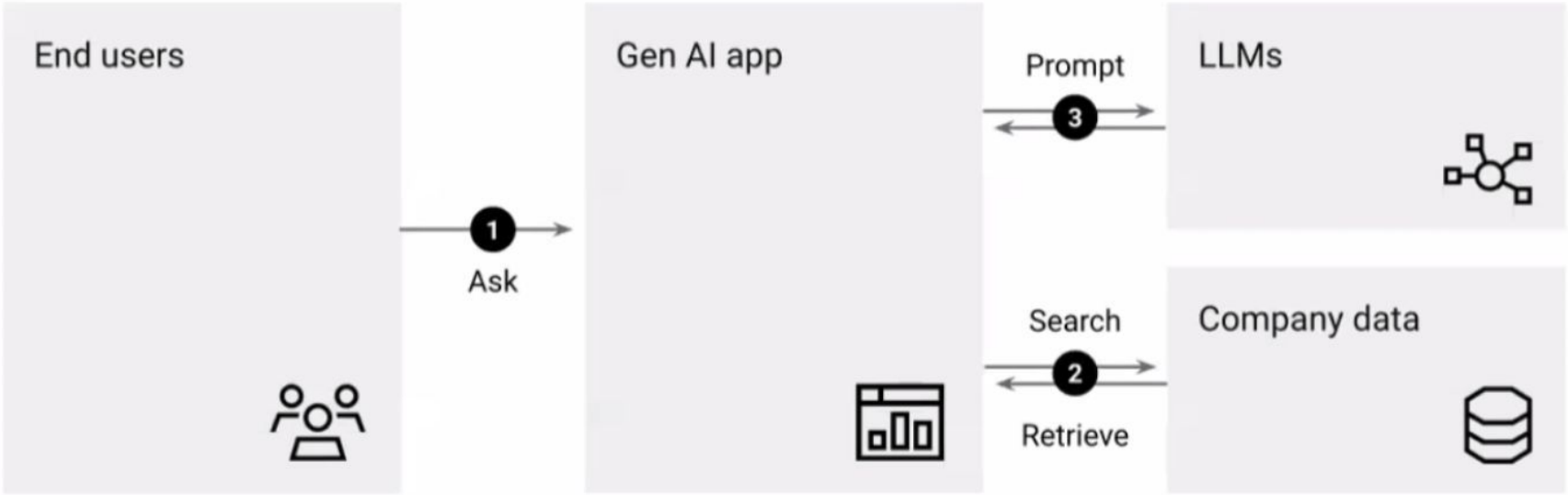
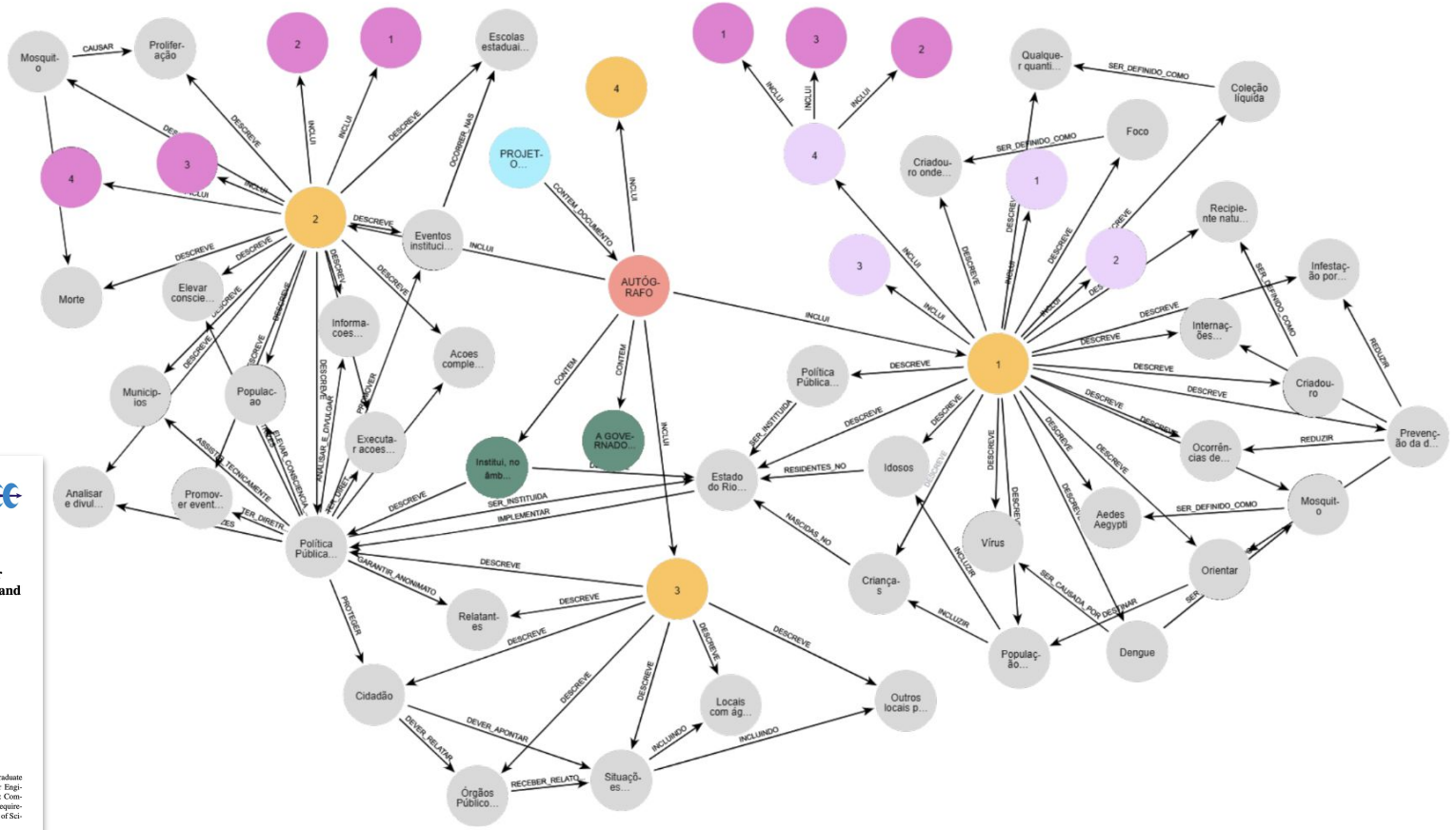


Figure 6.16: Contextually enriched KG of Bill of Law Nº 239/2019.



Source: Created by the author (2025).

UFRN Universidade Federal do Rio Grande do Norte
Centro de Tecnologia
Programa de Pós-Graduação em Engenharia Elétrica e de Computação

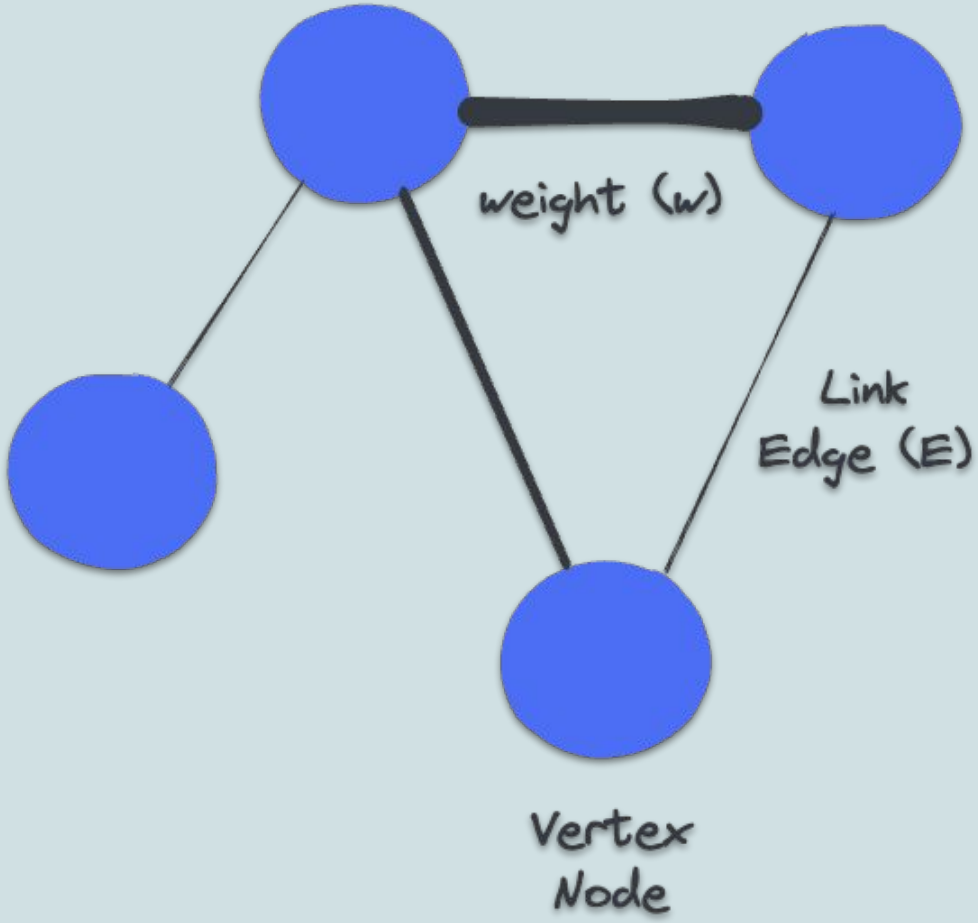
A Knowledge Graph-Based Approach for Modeling Legislative Texts: Representation and Document Similarity Analysis

Gisliany Lillian Alves de Oliveira

Advisor: Prof. Dr. Ivanovitch Medeiros Dantas da Silva

Doctoral Thesis presented to the Graduate Program of Electrical and Computer Engineering of UFRN (area of knowledge: Computing Engineering) as part of the requirements for obtaining the title of Doctor of Science.

PPgEEC Order Number: D392
Natal, RN, August 2025

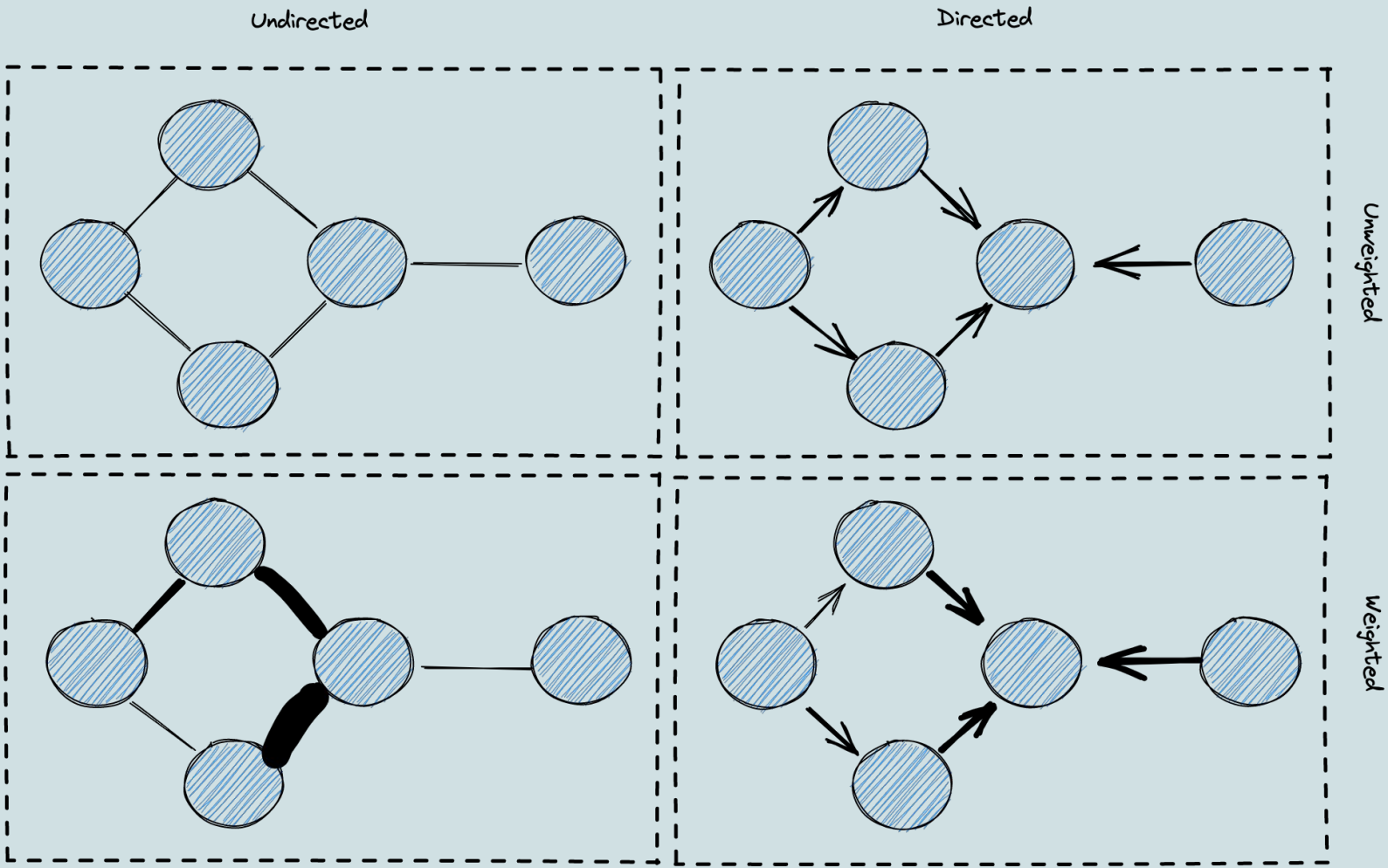


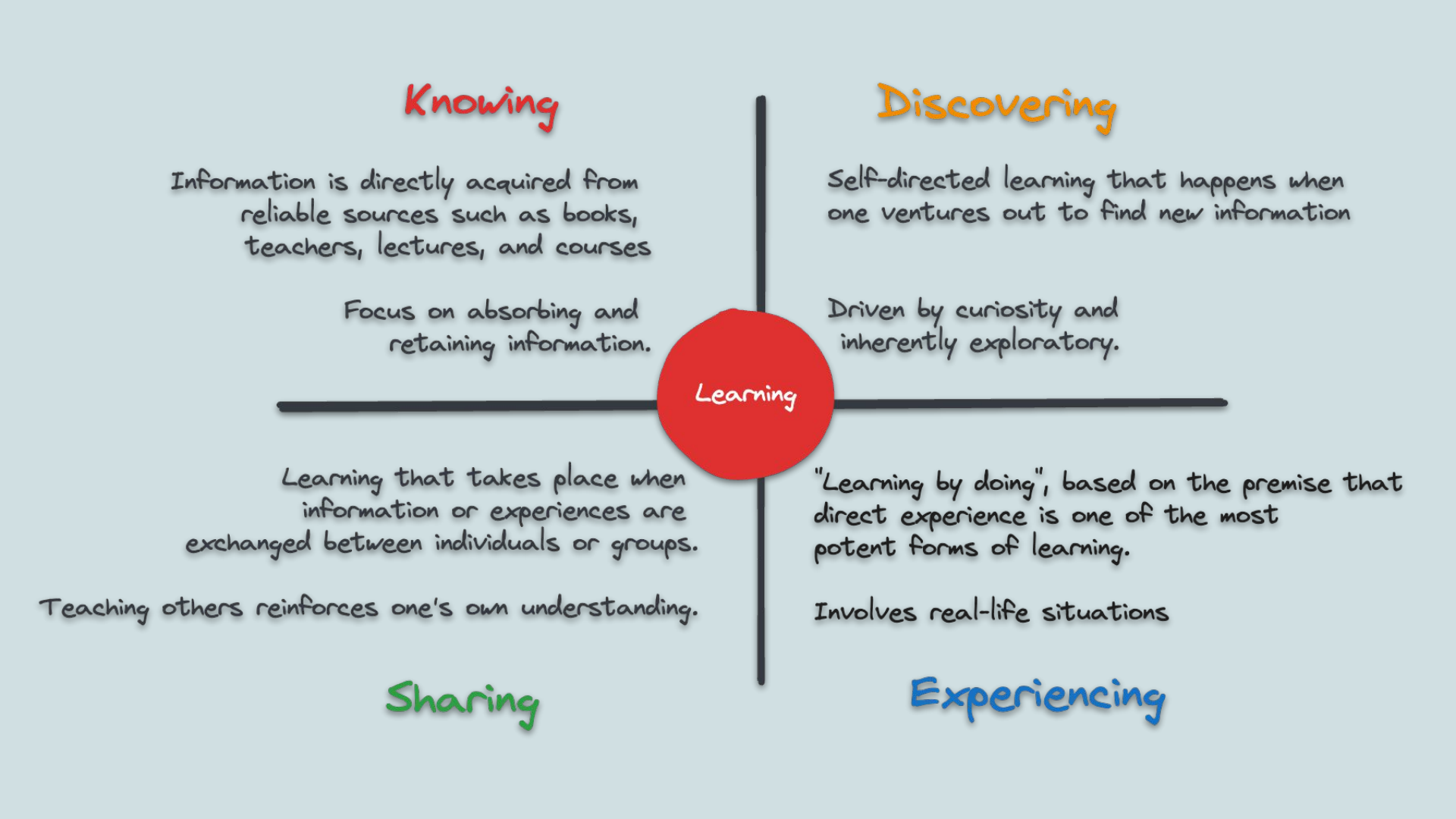
Basic Definitions

$$G = (V, E, W)$$

$$E \subseteq V \times V$$

$$W \subseteq R^{+}$$


Undirected
Directed
Unweighted
Weighted


Knowing
Information is directly acquired from reliable sources such as books, teachers, lectures, and courses
Focus on absorbing and retaining information.
Discovering
Self-directed learning that happens when one ventures out to find new information
Driven by curiosity and inherently exploratory.
Learning
Learning that takes place when information or experiences are exchanged between individuals or groups.
Teaching others reinforces one's own understanding.
Sharing
"Learning by doing", based on the premise that direct experience is one of the most potent forms of learning.
Involves real-life situations
Experiencing

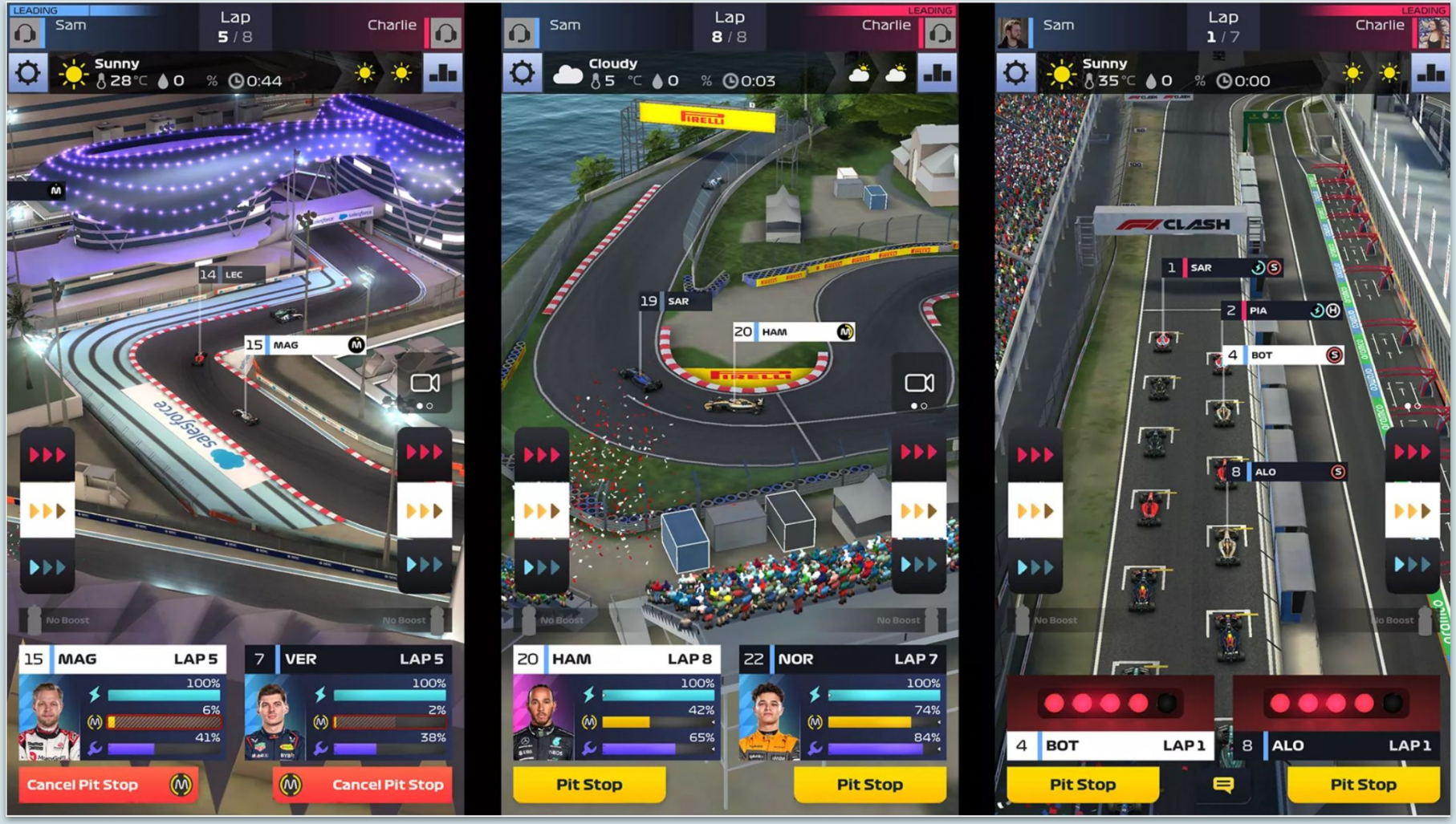
Hutch

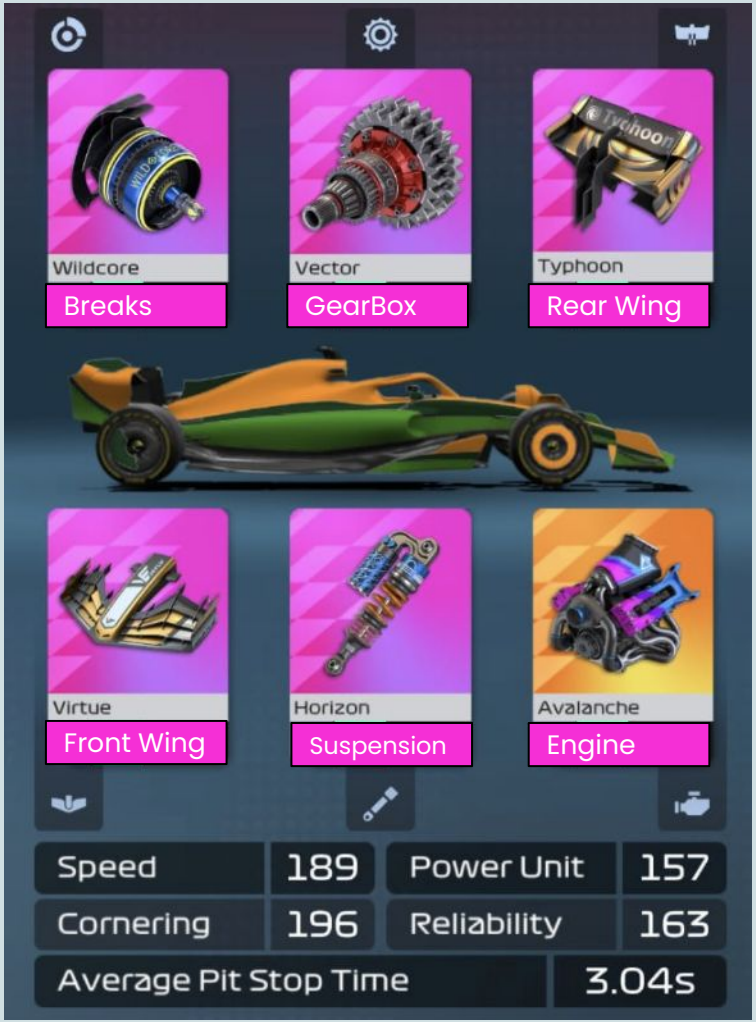
F1® CLASH

F1® CLASH

Compete in thrilling 1v1 races with the toughest rival race drivers from around the world.




LEADING
Sam
Lap 5 / 8
Charlie
Sunny
28 °C 0 % 0:44
14 LEC
15 MAG
No Boost
No Boost
15 MAG LAP 5
100%
6%
41%
Cancel Pit Stop
7 VER LAP 5
100%
2%
38%
Cancel Pit Stop
Sam
Lap 8 / 8
LEADING
Charlie
Cloudy
5 °C 0 % 0:03
19 SAR
20 HAM
No Boost
No Boost
20 HAM LAP 8
100%
42%
65%
Pit Stop
22 NOR LAP 7
100%
74%
84%
Pit Stop
Sam
Lap 1 / 7
LEADING
Charlie
Sunny
35 °C 0 % 0:00
F1 CLASH
1 SAR
2 PIA
4 BOT
8 ALO
No Boost
No Boost
4 BOT LAP 1
Pit Stop
8 ALO LAP 1
Pit Stop



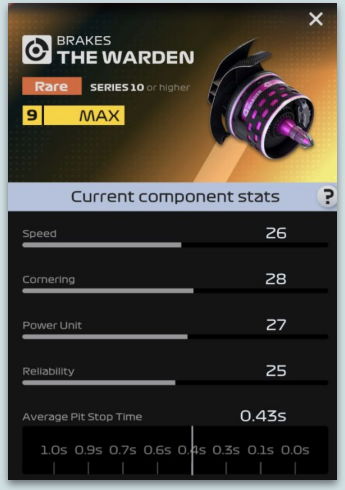
Configuring a setup

$\min(\lceil x \rceil, 220)$

Breaks


BRAKES
WILDCORE
Epic
SERIES 5 or higher
8 MAX
Current component stats
Speed 36
Cornering 23
Power Unit 33
Reliability 22
Average Pit Stop Time 0.59s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

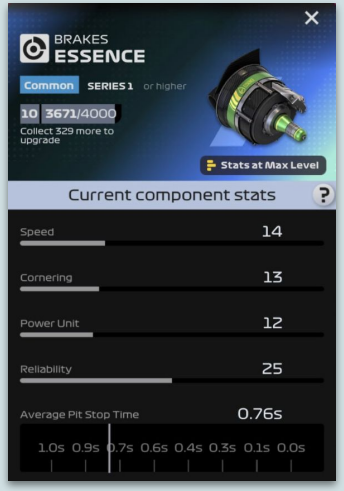

BRAKES
SUSPENSE
Rare
SERIES 8 or higher
9 MAX
Current component stats
Speed 20
Cornering 32
Power Unit 23
Reliability 21
Average Pit Stop Time 0.37s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


BRAKES
THE WARDEN
Rare
SERIES 10 or higher
9 MAX
Current component stats
Speed 26
Cornering 28
Power Unit 27
Reliability 25
Average Pit Stop Time 0.43s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


BRAKES
ONYX
Epic
SERIES 12
8 MAX
Current component stats
Speed 26
Cornering 23
Power Unit 25
Reliability 50
Average Pit Stop Time 0.49s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


BRAKES
AXIOM
Common
SERIES 3 or higher
11 MAX
Current component stats
Speed 14
Cornering 34
Power Unit 18
Reliability 15
Average Pit Stop Time 0.67s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


BRAKES
CRISIS SL
Rare
SERIES 2 or higher
9 MAX
Current component stats
Speed 27
Cornering 16
Power Unit 18
Reliability 19
Average Pit Stop Time 0.51s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

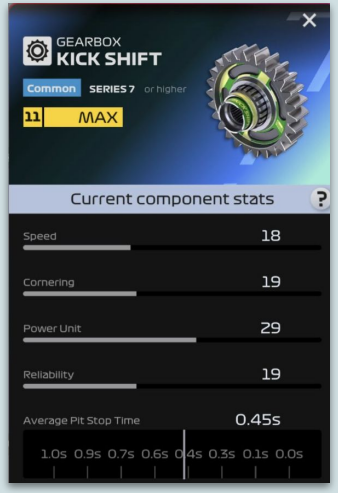

BRAKES
ESSENCE
Common
SERIES 1 or higher
10 3671/4000
Collect 329 more to upgrade
Stats at Max Level
Current component stats
Speed 14
Cornering 13
Power Unit 12
Reliability 25
Average Pit Stop Time 0.76s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


BRAKES
STARTER
Stock
Current component stats
Speed 1
Cornering 1
Power Unit 1
Reliability 1
Average Pit Stop Time 1s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

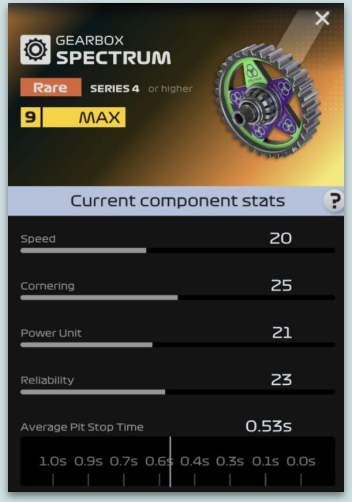
GearBox

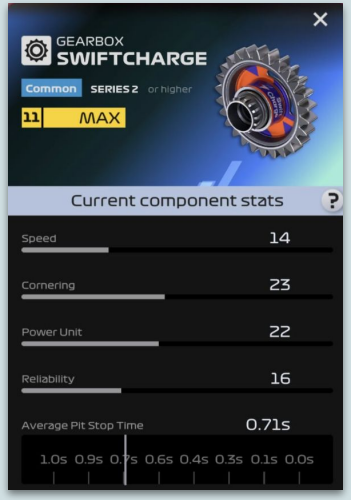

GEARBOX
VOYAGE
Epic
SERIES 12
8
MAX
Current component stats
Speed 23
Cornering 28
Power Unit 22
Reliability 27
Average Pit Stop Time 0s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

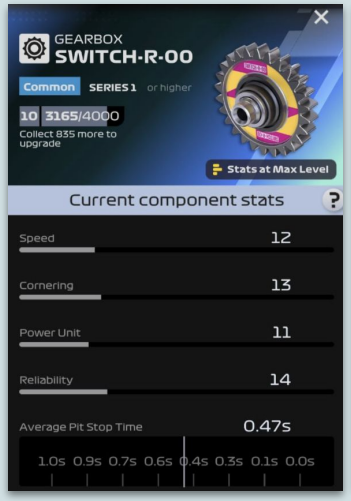

GEARBOX
VECTOR
Epic
SERIES 10 or higher
8
MAX
Current component stats
Speed 24
Cornering 38
Power Unit 22
Reliability 36
Average Pit Stop Time 0.55s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

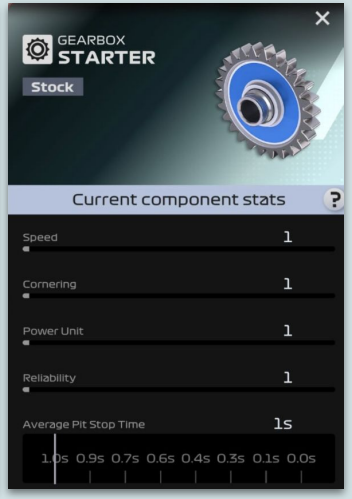

GEARBOX
KICK SHIFT
Common
SERIES 7 or higher
11
MAX
Current component stats
Speed 18
Cornering 19
Power Unit 29
Reliability 19
Average Pit Stop Time 0.45s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


GEARBOX
VERDICT
Rare
SERIES 6 or higher
9
MAX
Current component stats
Speed 33
Cornering 18
Power Unit 20
Reliability 30
Average Pit Stop Time 0.63s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

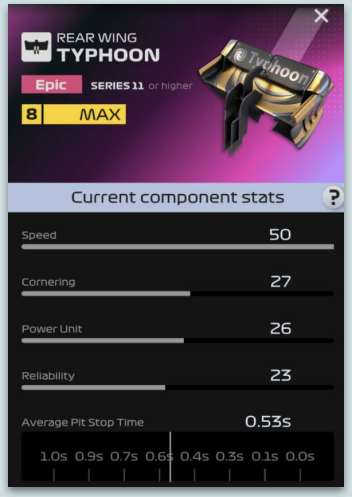

GEARBOX
SPECTRUM
Rare
SERIES 4 or higher
9
MAX
Current component stats
Speed 20
Cornering 25
Power Unit 21
Reliability 23
Average Pit Stop Time 0.53s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


GEARBOX
SWIFTCHARGE
Common
SERIES 2 or higher
11
MAX
Current component stats
Speed 14
Cornering 23
Power Unit 22
Reliability 16
Average Pit Stop Time 0.71s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


GEARBOX
SWITCH-R-00
Common
SERIES 1 or higher
10
3165/4000
Collect 835 more to upgrade
Stats at Max Level
Current component stats
Speed 12
Cornering 13
Power Unit 11
Reliability 14
Average Pit Stop Time 0.47s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

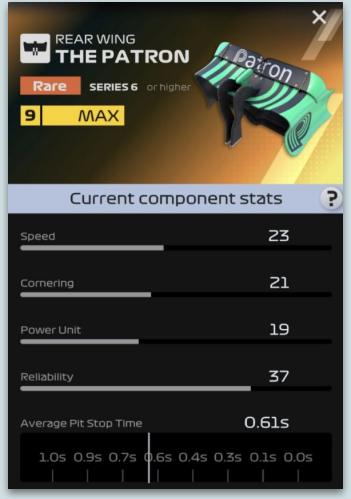

GEARBOX
STARTER
Stock
Current component stats
Speed 1
Cornering 1
Power Unit 1
Reliability 1
Average Pit Stop Time 1s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

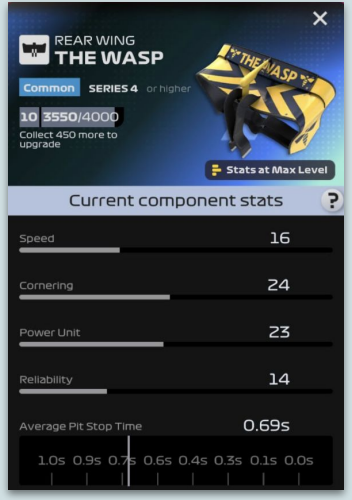
Rear Wing

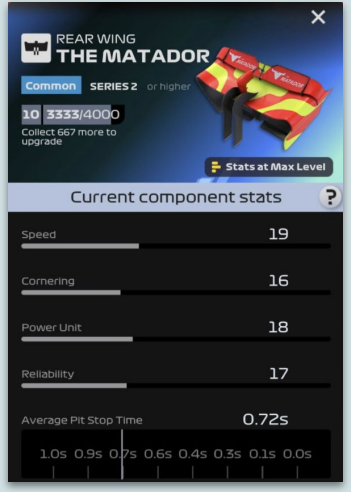

REAR WING
TYPHOON
Epic
SERIES 11 or higher
8 MAX
Current component stats
Speed 50
Cornering 27
Power Unit 26
Reliability 23
Average Pit Stop Time 0.53s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

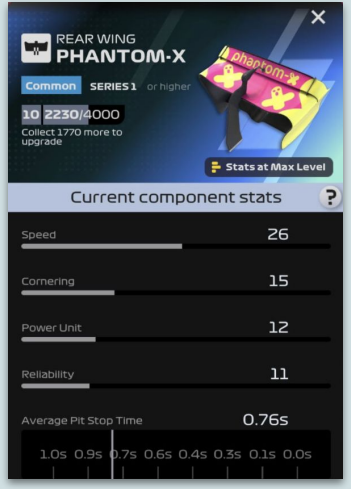

REAR WING
TRANSCENDENCE
Epic
SERIES 9 or higher
8 MAX
Current component stats
Speed 24
Cornering 22
Power Unit 36
Reliability 37
Average Pit Stop Time 0.53s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

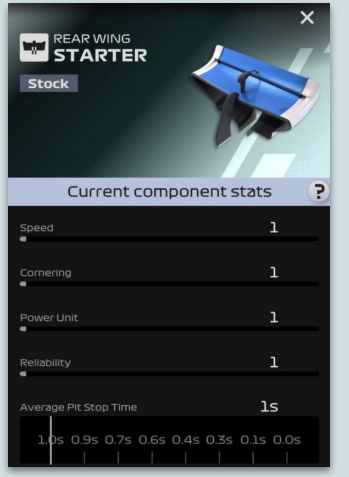

REAR WING
FREEFLARE
Rare
SERIES 8 or higher
9 MAX
Current component stats
Speed 21
Cornering 33
Power Unit 20
Reliability 22
Average Pit Stop Time 0.37s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


REAR WING
THE PATRON
Rare
SERIES 6 or higher
9 MAX
Current component stats
Speed 23
Cornering 21
Power Unit 19
Reliability 37
Average Pit Stop Time 0.61s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

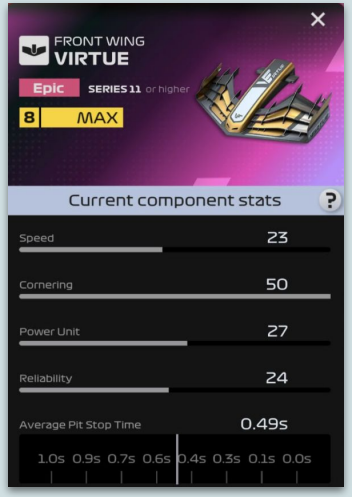

REAR WING
THE WASP
Common
SERIES 4 or higher
10 3550/4000
Collect 450 more to upgrade
Stats at Max Level
Current component stats
Speed 16
Cornering 24
Power Unit 23
Reliability 14
Average Pit Stop Time 0.69s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

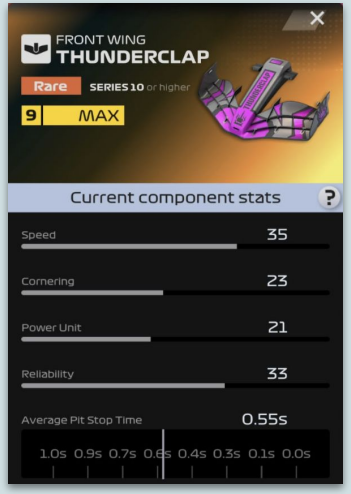

REAR WING
THE MATADOR
Common
SERIES 2 or higher
10 3333/4000
Collect 667 more to upgrade
Stats at Max Level
Current component stats
Speed 19
Cornering 16
Power Unit 18
Reliability 17
Average Pit Stop Time 0.72s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

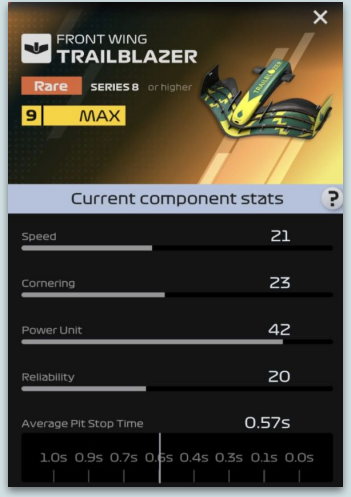

REAR WING
PHANTOM-X
Common
SERIES 1 or higher
10 2230/4000
Collect 1770 more to upgrade
Stats at Max Level
Current component stats
Speed 26
Cornering 15
Power Unit 12
Reliability 11
Average Pit Stop Time 0.76s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

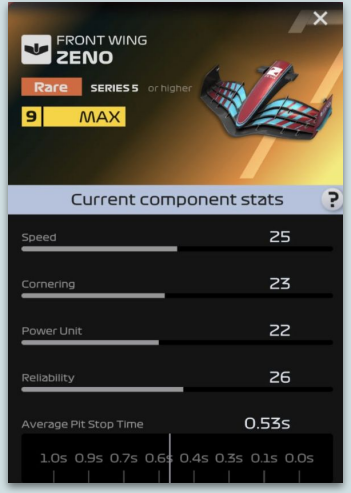

REAR WING
STARTER
Stock
Current component stats
Speed 1
Cornering 1
Power Unit 1
Reliability 1
Average Pit Stop Time 1s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

Front Wing

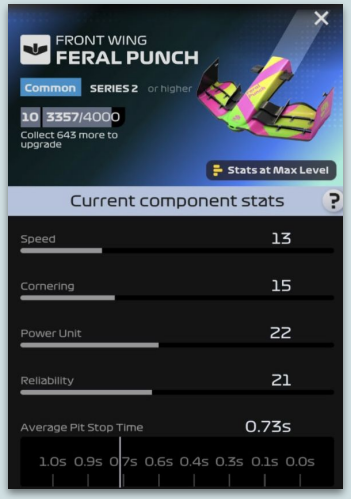

FRONT WING
VIRTUE
Epic SERIES 11 or higher
8 MAX
Current component stats
Speed 23
Cornering 50
Power Unit 27
Reliability 24
Average Pit Stop Time 0.49s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

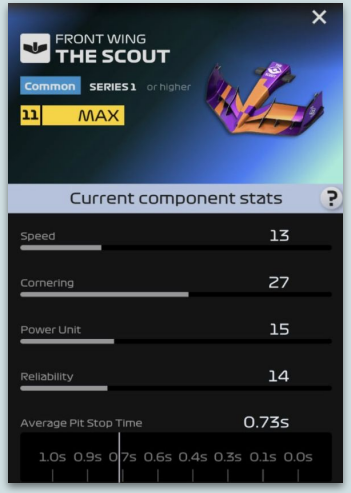

FRONT WING
THUNDERCLAP
Rare SERIES 10 or higher
9 MAX
Current component stats
Speed 35
Cornering 23
Power Unit 21
Reliability 33
Average Pit Stop Time 0.55s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

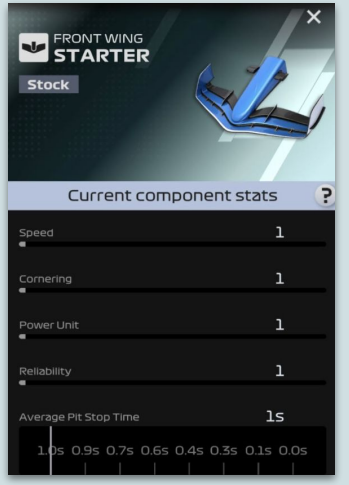

FRONT WING
TRAILBLAZER
Rare SERIES 8 or higher
9 MAX
Current component stats
Speed 21
Cornering 23
Power Unit 42
Reliability 20
Average Pit Stop Time 0.57s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


FRONT WING
ZENO
Rare SERIES 5 or higher
9 MAX
Current component stats
Speed 25
Cornering 23
Power Unit 22
Reliability 26
Average Pit Stop Time 0.53s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

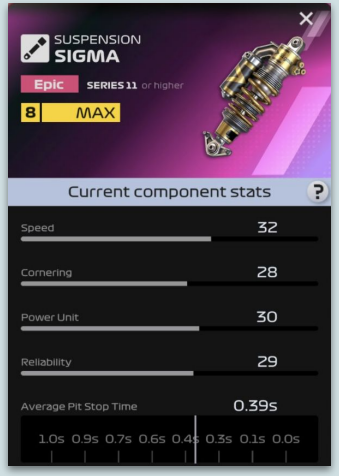

FRONT WING
THE VAGABOND
Epic SERIES 3 or higher
8 MAX
Current component stats
Speed 31
Cornering 20
Power Unit 23
Reliability 21
Average Pit Stop Time 0.35s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

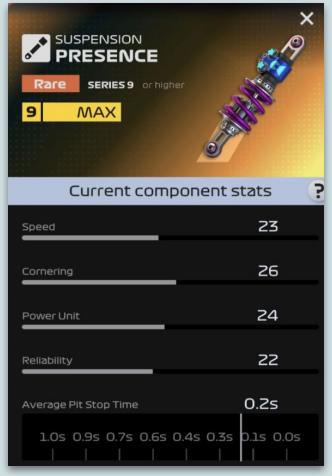

FRONT WING
FERAL PUNCH
Common SERIES 2 or higher
10 3357/4000
Collect 643 more to upgrade
Stats at Max Level
Current component stats
Speed 13
Cornering 15
Power Unit 22
Reliability 21
Average Pit Stop Time 0.73s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

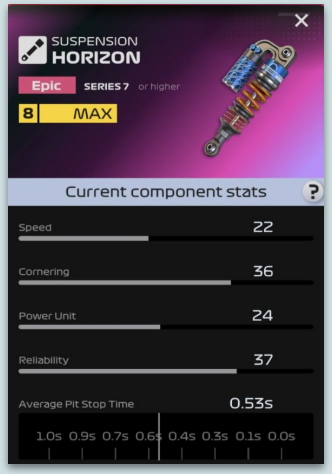

FRONT WING
THE SCOUT
Common SERIES 1 or higher
11 MAX
Current component stats
Speed 13
Cornering 27
Power Unit 15
Reliability 14
Average Pit Stop Time 0.73s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


FRONT WING
STARTER
Stock
Current component stats
Speed 1
Cornering 1
Power Unit 1
Reliability 1
Average Pit Stop Time 1s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

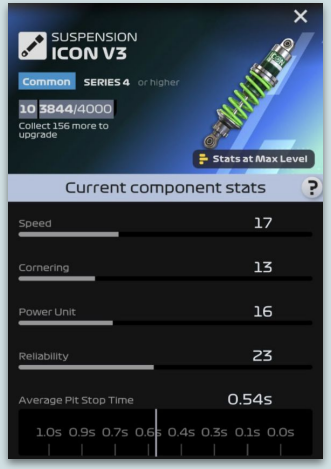
Suspension

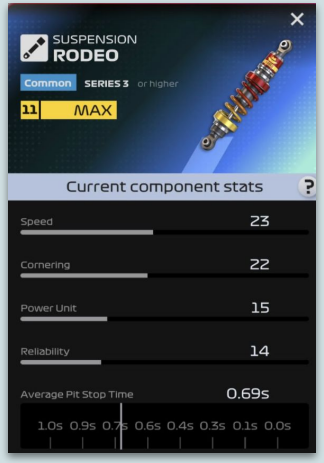

SUSPENSION
SIGMA
Epic
SERIES 11 or higher
8 MAX
Current component stats
Speed 32
Cornering 28
Power Unit 30
Reliability 29
Average Pit Stop Time 0.39s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

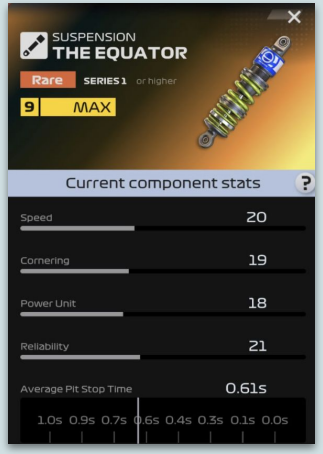

SUSPENSION
PRESENCE
Rare
SERIES 9 or higher
9 MAX
Current component stats
Speed 23
Cornering 26
Power Unit 24
Reliability 22
Average Pit Stop Time 0.2s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

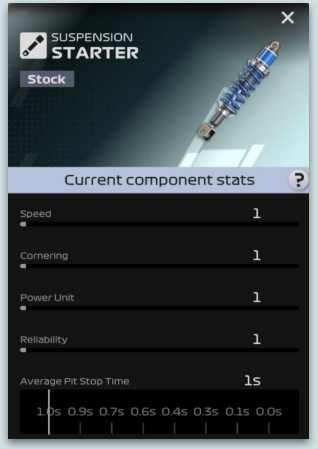

SUSPENSION
HORIZON
Epic
SERIES 7 or higher
8 MAX
Current component stats
Speed 22
Cornering 36
Power Unit 24
Reliability 37
Average Pit Stop Time 0.53s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


SUSPENSION
RADIANCE
Common
SERIES 6 or higher
11 MAX
Current component stats
Speed 25
Cornering 17
Power Unit 26
Reliability 19
Average Pit Stop Time 0.65s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


SUSPENSION
ICON V3
Common
SERIES 4 or higher
10 3844/4000
Collect 156 more to upgrade
Stats at Max Level
Current component stats
Speed 17
Cornering 13
Power Unit 16
Reliability 23
Average Pit Stop Time 0.54s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


SUSPENSION
RODEO
Common
SERIES 3 or higher
11 MAX
Current component stats
Speed 23
Cornering 22
Power Unit 15
Reliability 14
Average Pit Stop Time 0.69s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


SUSPENSION
THE EQUATOR
Rare
SERIES 1 or higher
9 MAX
Current component stats
Speed 20
Cornering 19
Power Unit 18
Reliability 21
Average Pit Stop Time 0.61s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


SUSPENSION
STARTER
Stock
Current component stats
Speed 1
Cornering 1
Power Unit 1
Reliability 1
Average Pit Stop Time 1s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

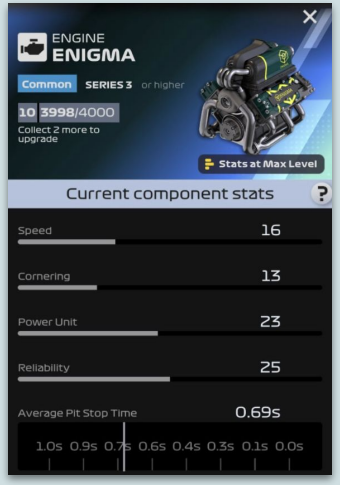
Engine


ENGINE
CLOUDROAR
Epic
SERIES 12
8
MAX
Current component stats
Speed 26
Cornering 24
Power Unit 50
Reliability 27
Average Pit Stop Time 0.55s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


ENGINE
AVALANCHE
Rare
SERIES 9 or higher
9
MAX
Current component stats
Speed 34
Cornering 22
Power Unit 25
Reliability 21
Average Pit Stop Time 0.35s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

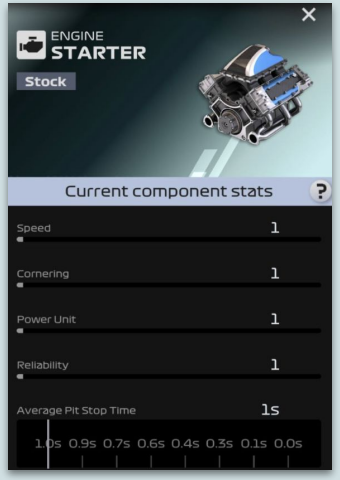

ENGINE
THE ROVER
Rare
SERIES 7 or higher
9
MAX
Current component stats
Speed 27
Cornering 25
Power Unit 28
Reliability 24
Average Pit Stop Time 0.53s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

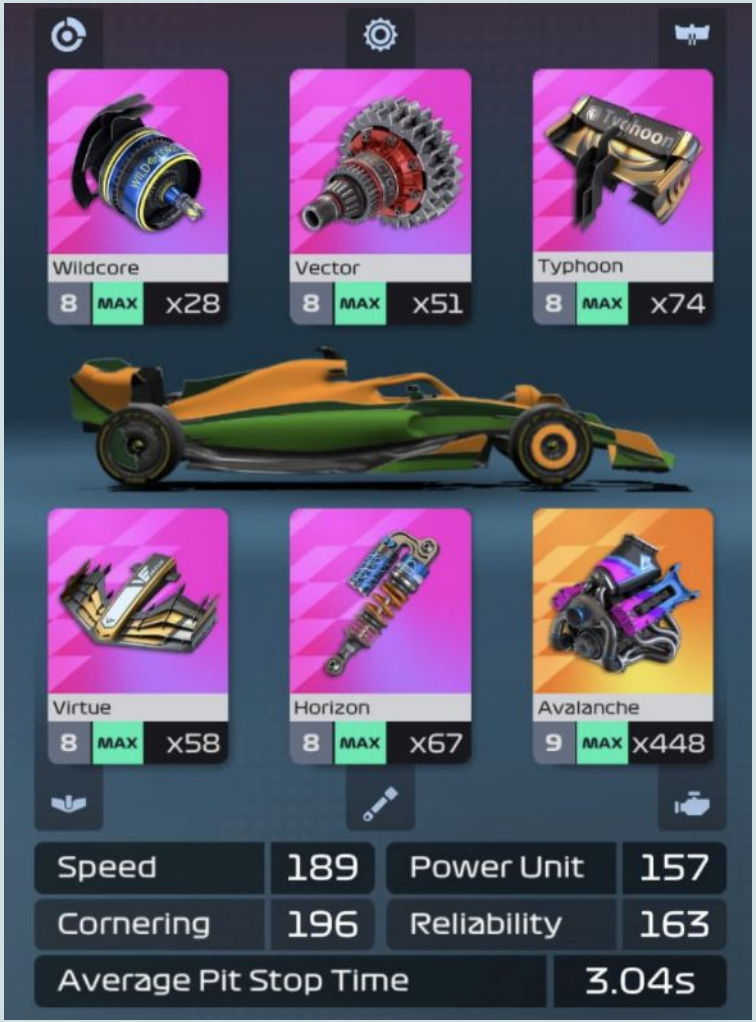

ENGINE
TWINBURST
Common
SERIES 5 or higher
11
MAX
Current component stats
Speed 16
Cornering 29
Power Unit 18
Reliability 17
Average Pit Stop Time 0.51s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

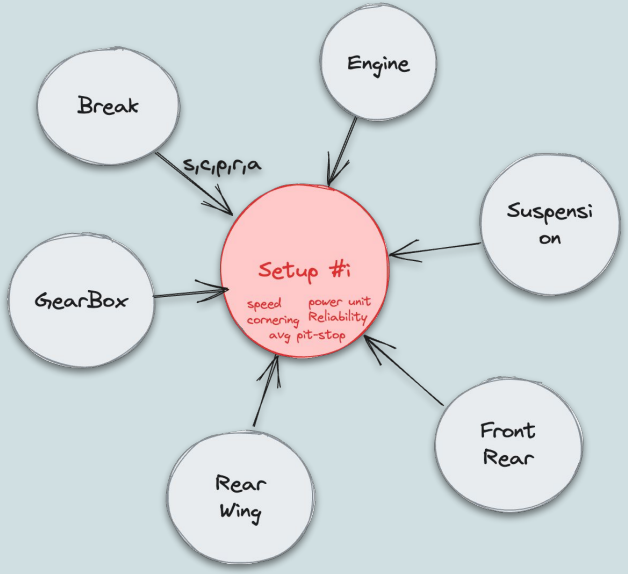

ENGINE
ENIGMA
Common
SERIES 3 or higher
10 3998/4000
Collect 2 more to upgrade
Stats at Max Level
Current component stats
Speed 16
Cornering 13
Power Unit 23
Reliability 25
Average Pit Stop Time 0.69s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

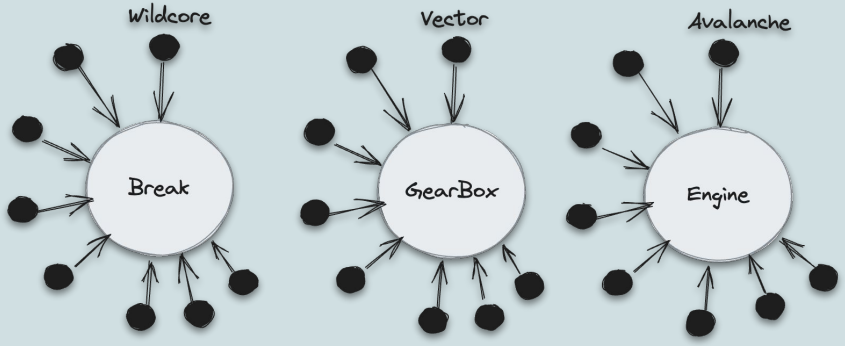

ENGINE
NOVA
Common
SERIES 2 or higher
11
MAX
Current component stats
Speed 31
Cornering 13
Power Unit 15
Reliability 16
Average Pit Stop Time 0.71s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

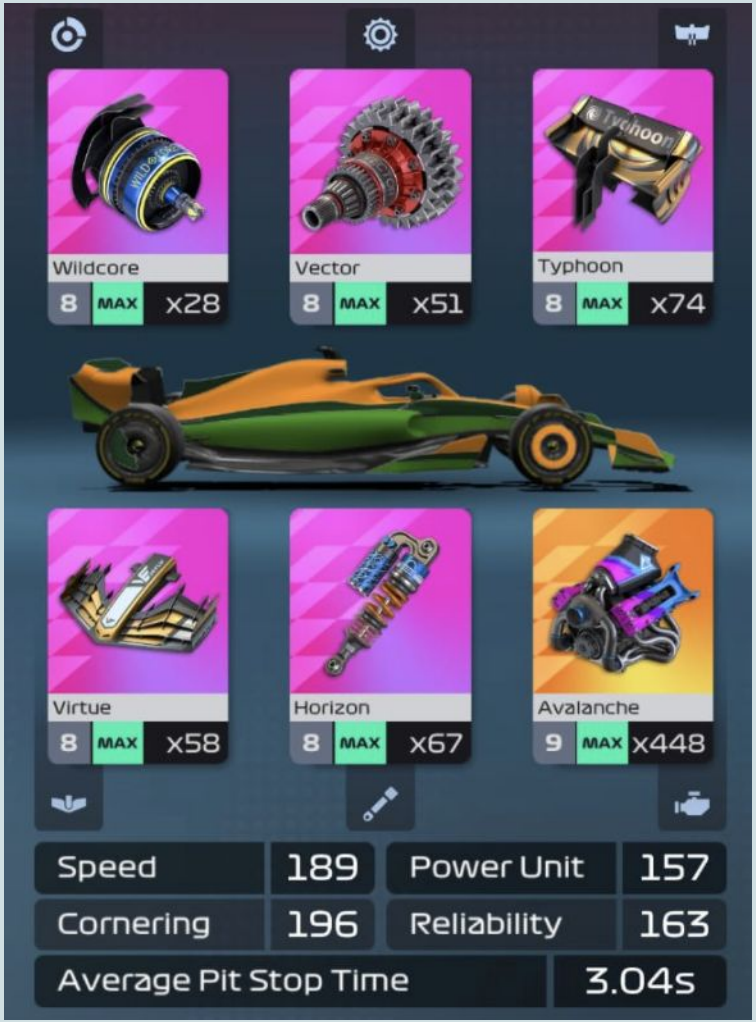

ENGINE
BRUTE FORCE
Epic
SERIES 1 or higher
8
MAX
Current component stats
Speed 21
Cornering 19
Power Unit 36
Reliability 18
Average Pit Stop Time 0.63s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s


ENGINE
STARTER
Stock
Current component stats
Speed 1
Cornering 1
Power Unit 1
Reliability 1
Average Pit Stop Time 1s
1.0s 0.9s 0.7s 0.6s 0.4s 0.3s 0.1s 0.0s

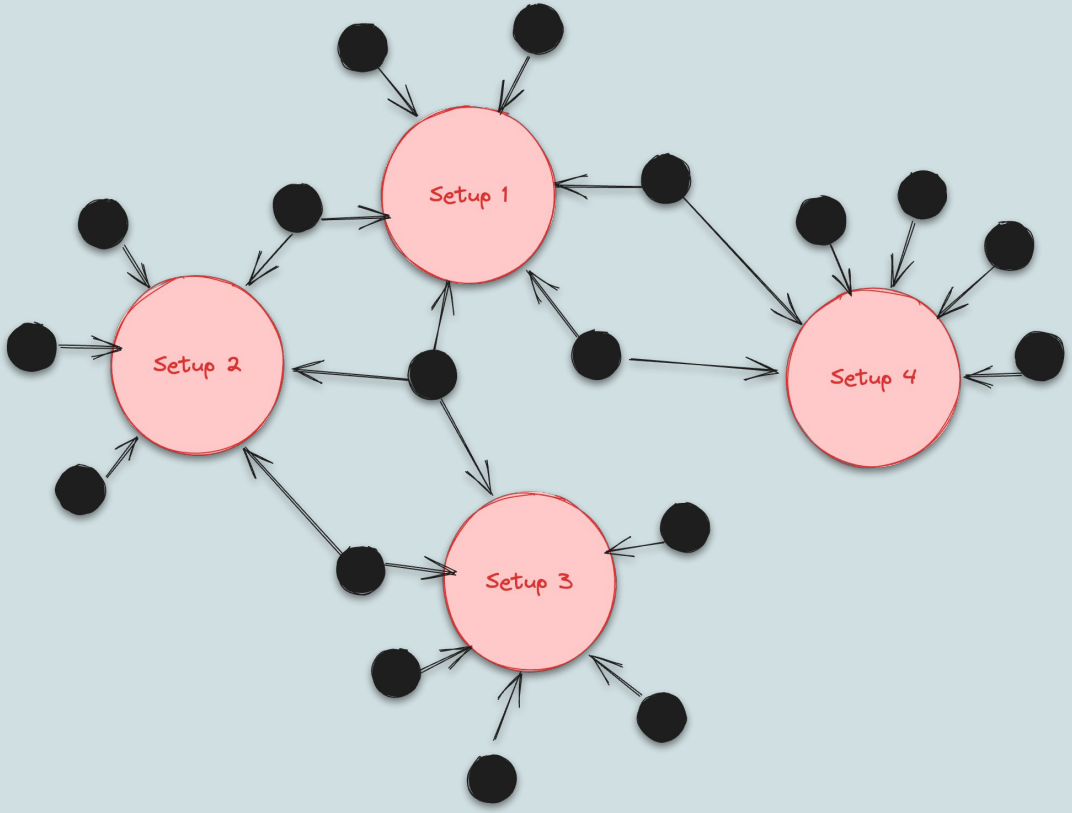

Wildcore
8 MAX x28
Vector
8 MAX x51
Typhoon
8 MAX x74
Virtue
8 MAX x58
Horizon
8 MAX x67
Avalanche
9 MAX x448
Speed 189
Power Unit 157
Cornering 196
Reliability 163
Average Pit Stop Time 3.04s


Break
Engine
Suspensi on
GearBox
Setup #i
speed power unit
cornering Reliabilty
avg pit-stop
Front Rear
Rear Wing


Wildcore
Break
Vector
GearBox
Avalanche
Engine



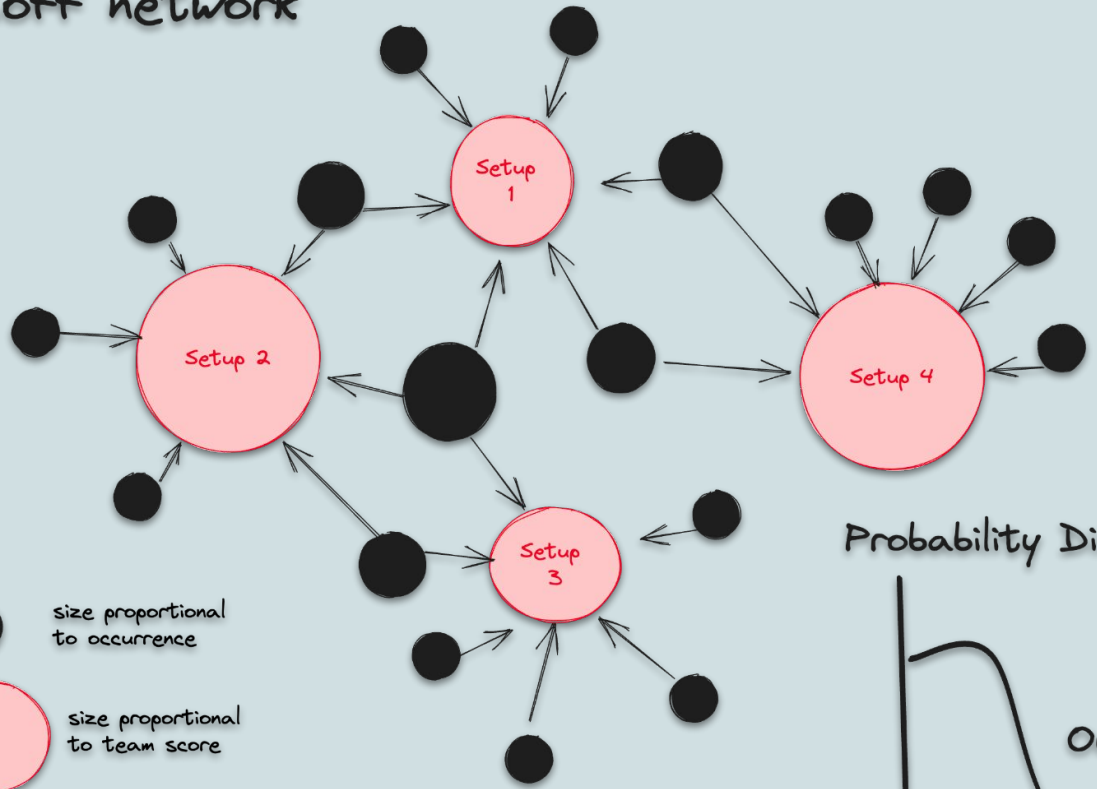
$8^6 = 262{,}144$ combinations



Team Score = speed + cornering + power unit + reliability +
(avg pit stop)/0.02



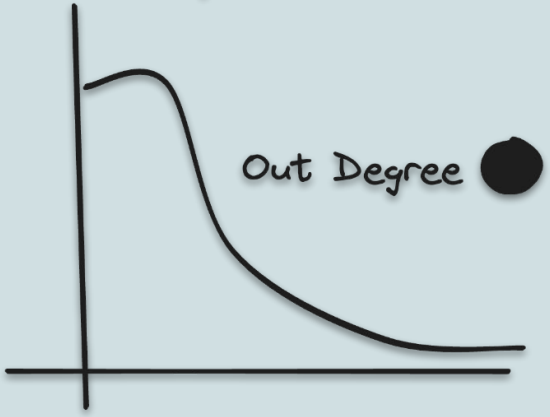
Cutoff network



size proportional to occurrence

size proportional to team score

Probability Distribution Function (PDF)




+10
+25
+15
Tsar


+25
+15
+10
Frost


+10
+20
+20
Tulip


+20
+15
+15
Dragon


+15
+20
+15
Kawaii


+25
+15
+10
Pretzel


+10
+15
+25
Vice


+10
+25
+15
Schooner


+15
+15
+20
Djinn


+15
+10
+25
Oud


+25
+10
+15
Eternal Flame


+20
+15
+15
Eagle


+10
+20
+20
Iron Force


+15
+10
+25
Lumberjack


+20
+20
+10
Cranberry


+20
+25
+5
Butterfly


+10
+15
+25
Tune-in


+5
+5
+5
Self-Control


+5
+5
+10
Warrior


+5
+10
+10
Ballast

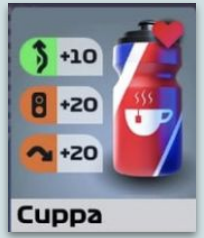

+10
+15
+5
Instinct


+15
+15
+5
Downforce


+20
+15
+5
H
Hex


+15
+10
+25
Eggception


+20
+20
+10
Rooster


+10
+20
+20
Cuppa


+10
+25
+15
Street Shark


+10
+25
+15
Herald


+10
+20
+20
Prince


+25
+15
+10
Unstoppable

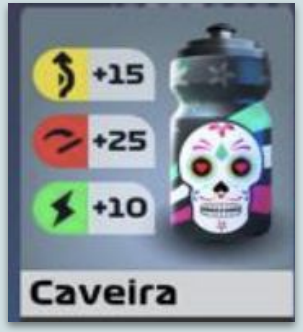

+5
+25
+20
Dead Fast


+25
+15
+10
Gladiator

















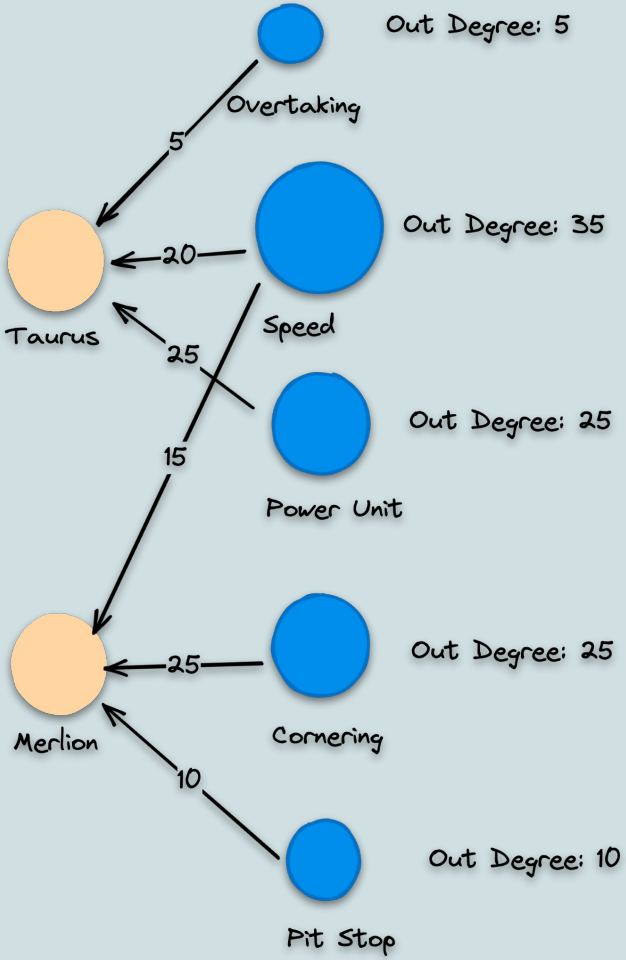




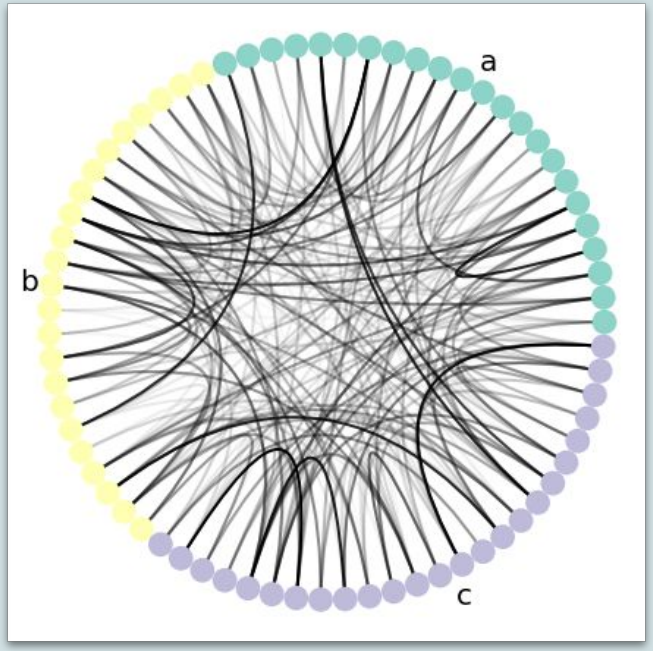




Speed, Cornering, Power_Unit, Reliability, Pit_Stop, Overtaking, Defending, Race_Start, Tyre_Management





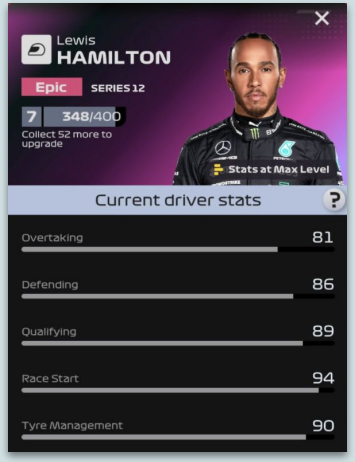


https://ericmjl.github.io/nxviz/

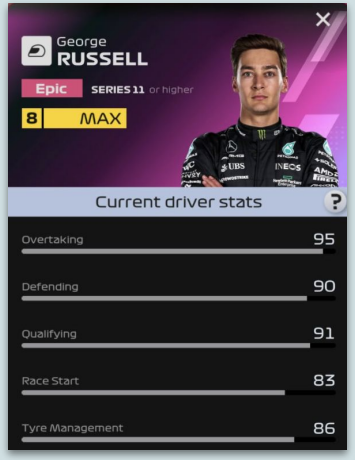


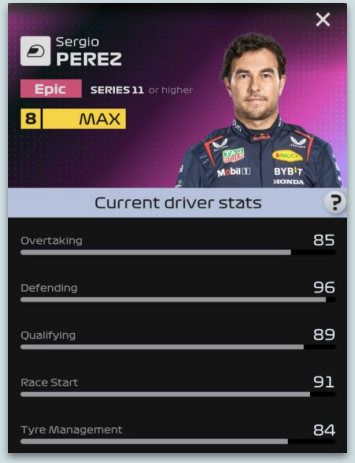




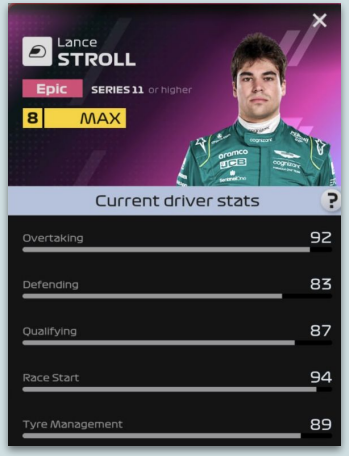


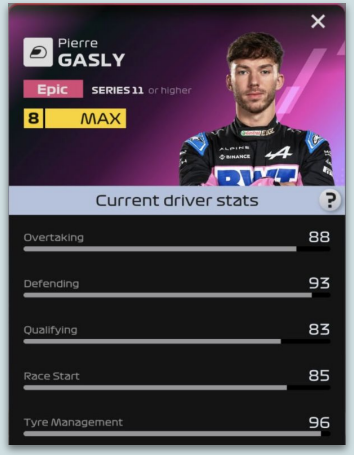




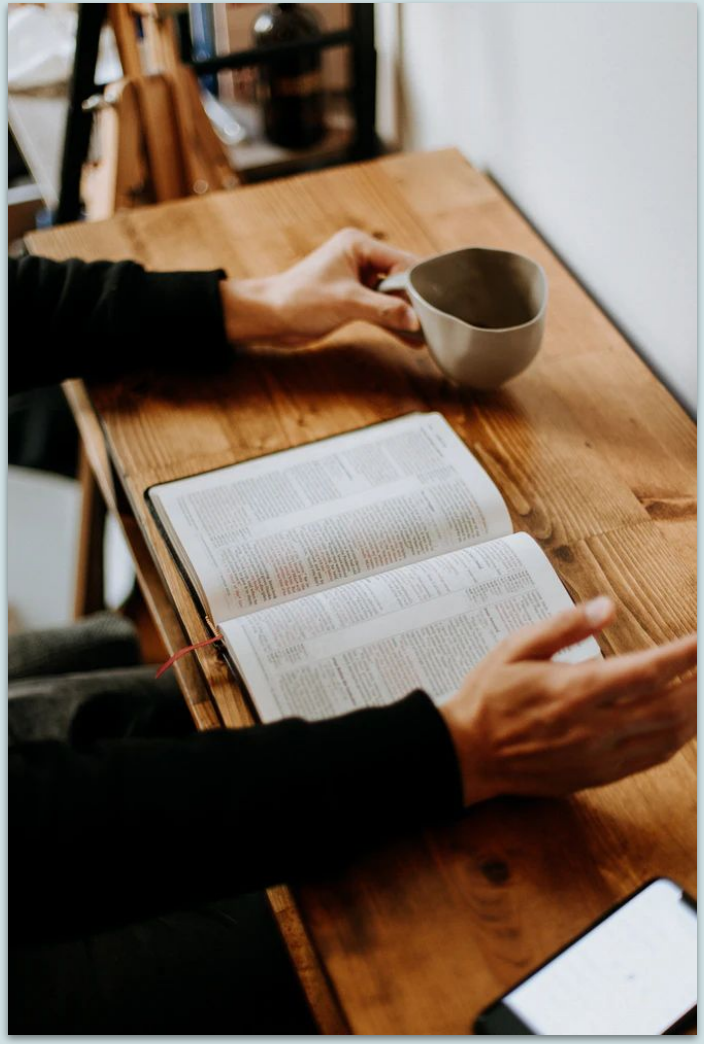








$\min(\lceil x1,99)$



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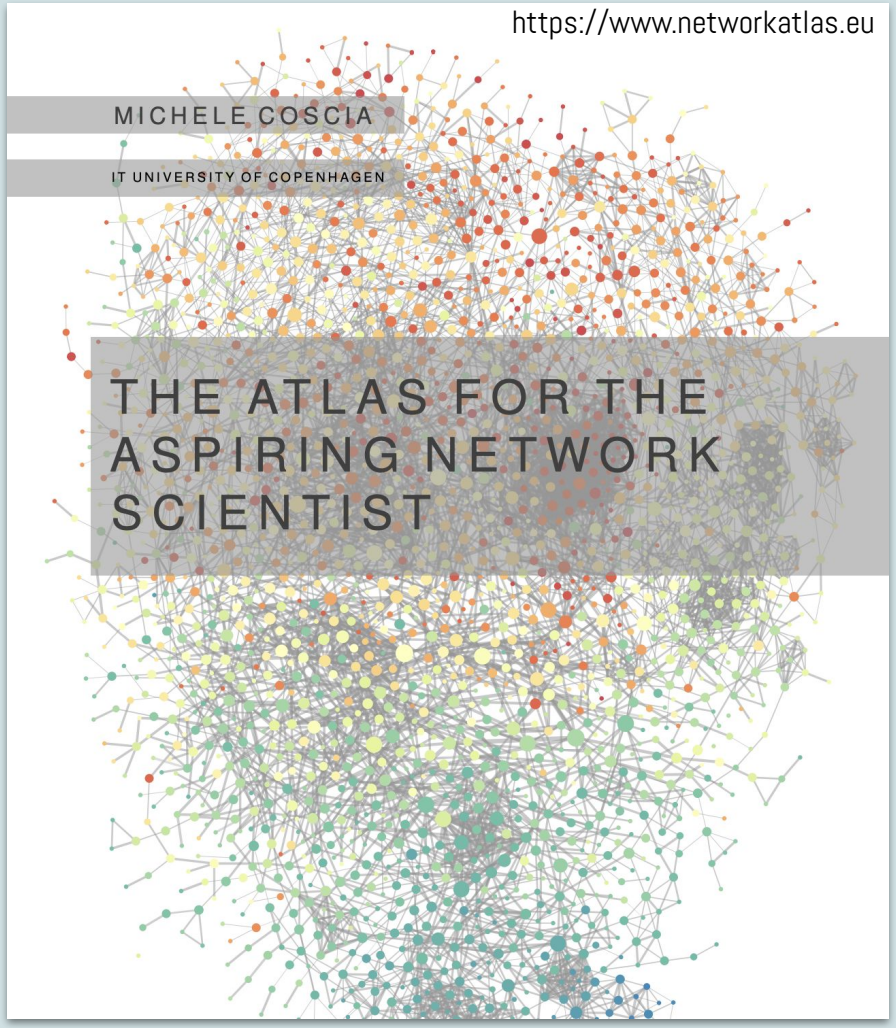
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Contact

Mailing list
Issue tracker
Source

Releases

Stable (notes)

3.6.1 — Dec 2025
Documentation

Latest (notes)

3.7 development
Documentation

Archive

Language

English

NetworkX

Network Analysis in Python

NetworkX is a Python package for the creation, manipulation, and study of the structure, dynamics, and functions of complex networks.

Software for complex networks

- Data structures for graphs, digraphs, and multigraphs
- Many standard graph algorithms
- Network structure and analysis measures
- Generators for classic graphs, random graphs, and synthetic networks
- Nodes can be "anything" (e.g., text, images, XML records)
- Edges can hold arbitrary data (e.g., weights, time-series)
- Open source 3-clause BSD license
- Well tested with over 90% code coverage
- Additional benefits from Python include fast prototyping, easy to teach, and multi-platform

https://networkx.org/

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